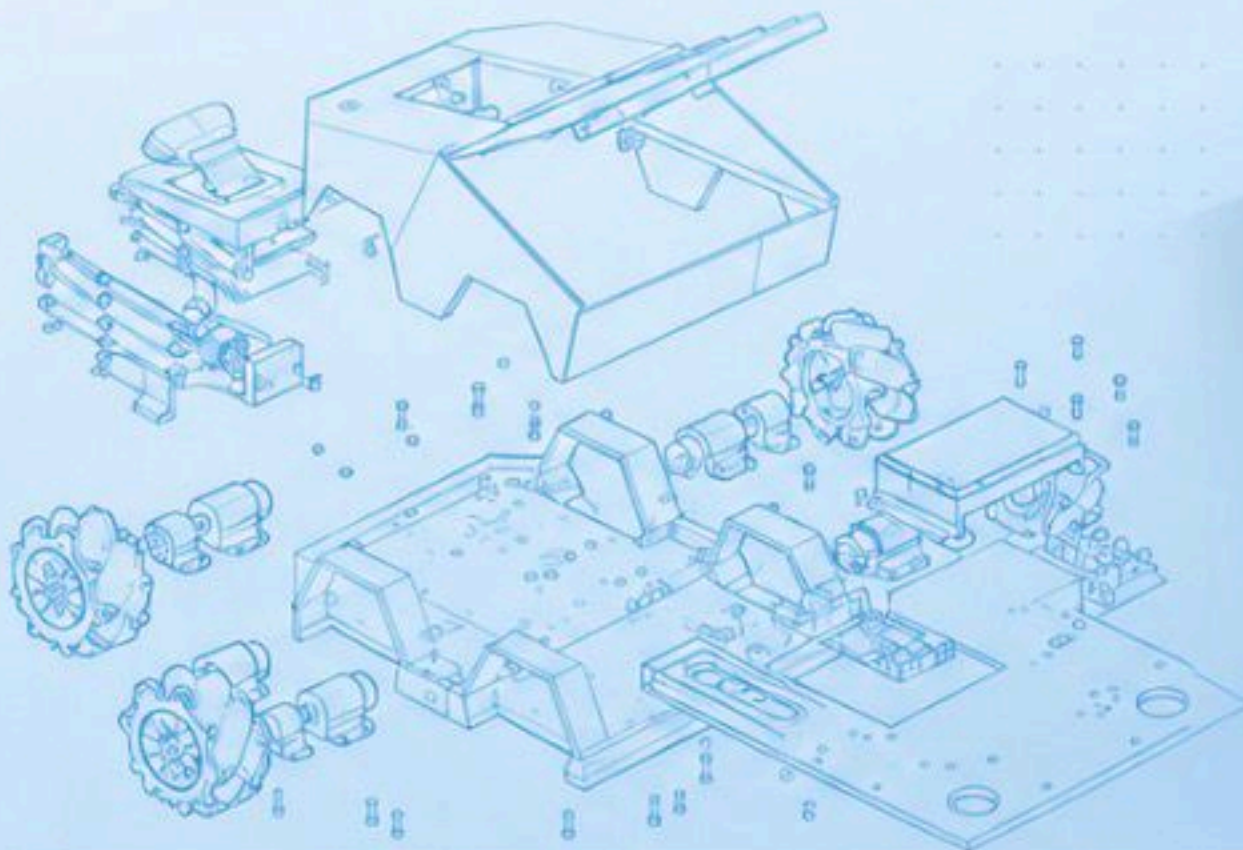
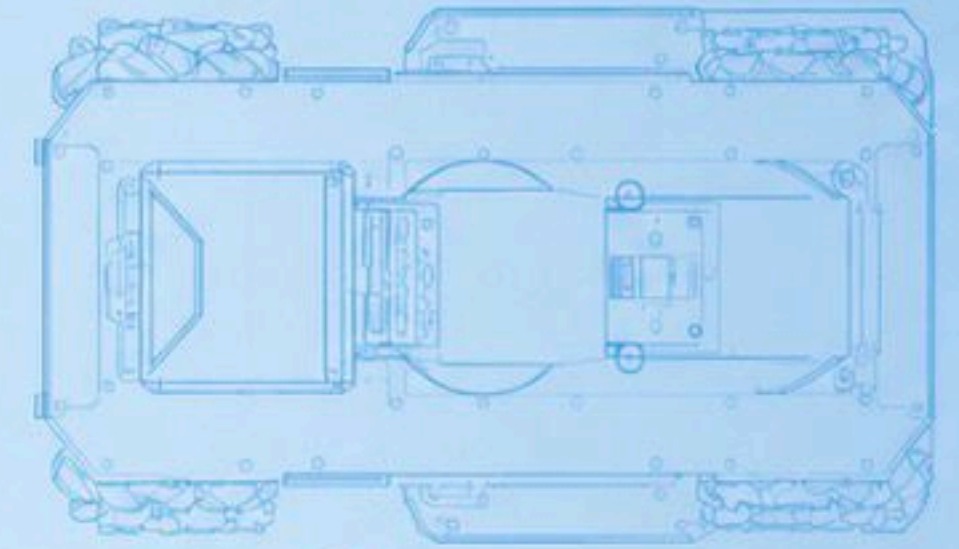
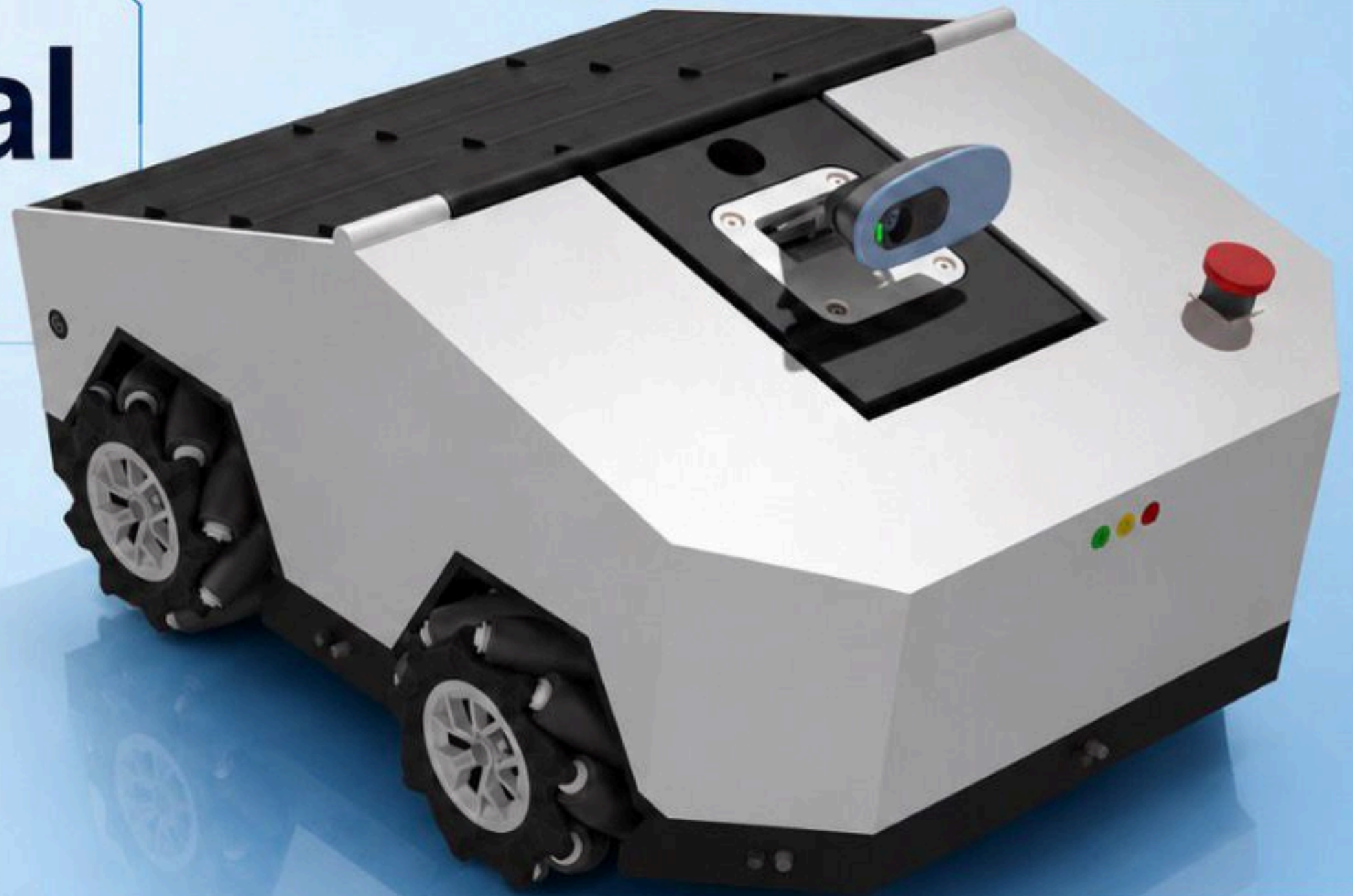


AUDIX

SCAN. VERIFY. REPORT.

User Manual

Autonomous Warehouse
Inventory Auditing Robot



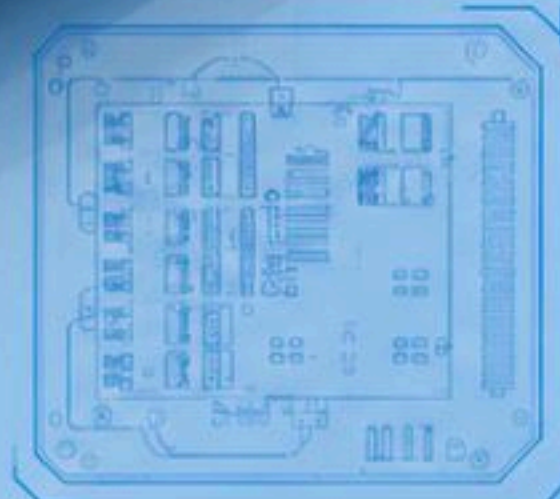
MCT332 / MCT334
– Design of Mechatronics 2



Spring 2026



Ain Shams University
– Faculty of Engineering



Document Information

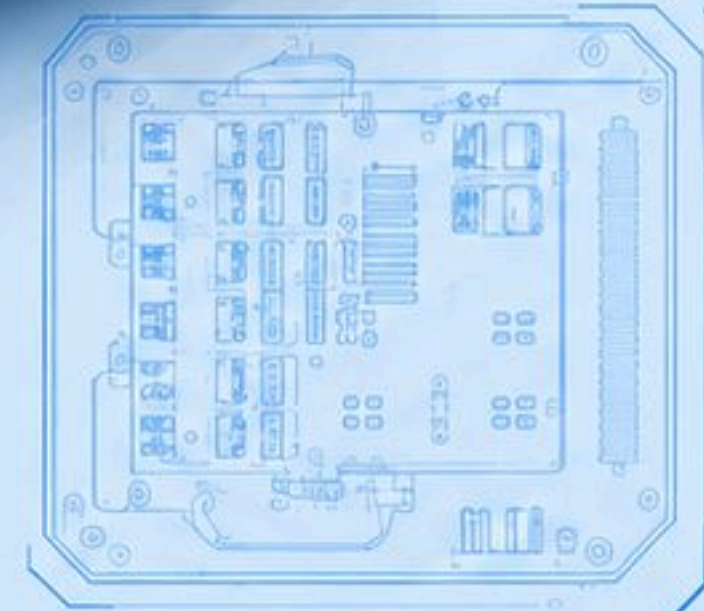
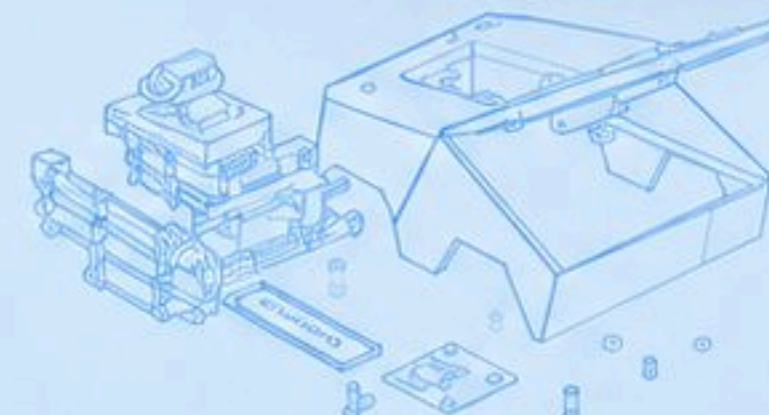
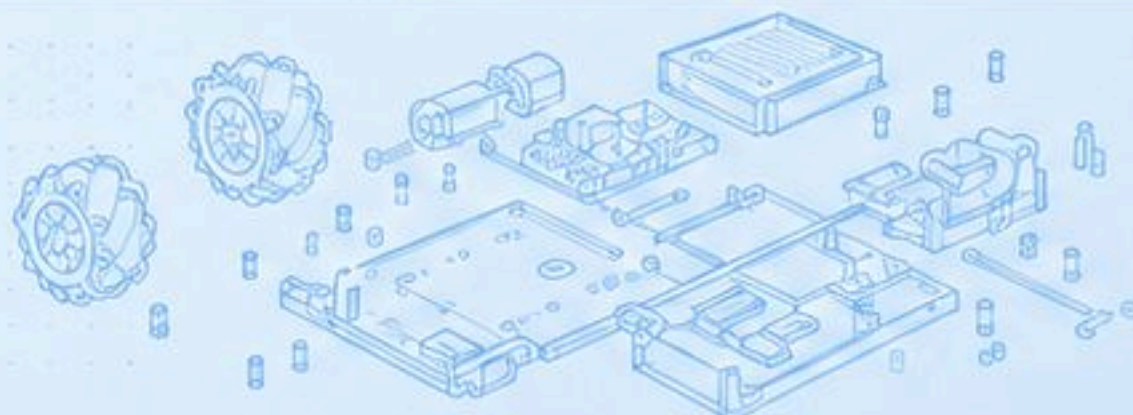
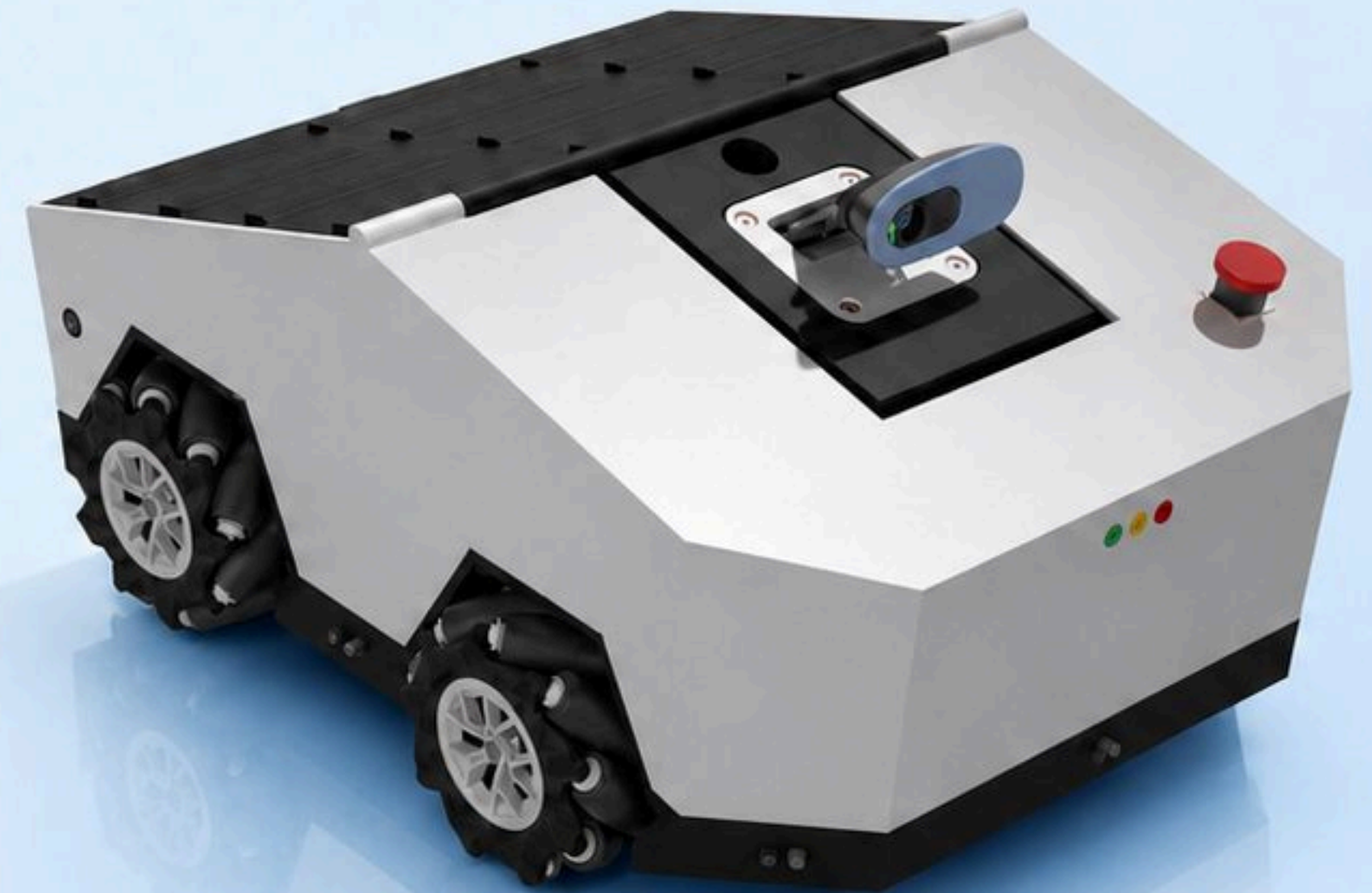
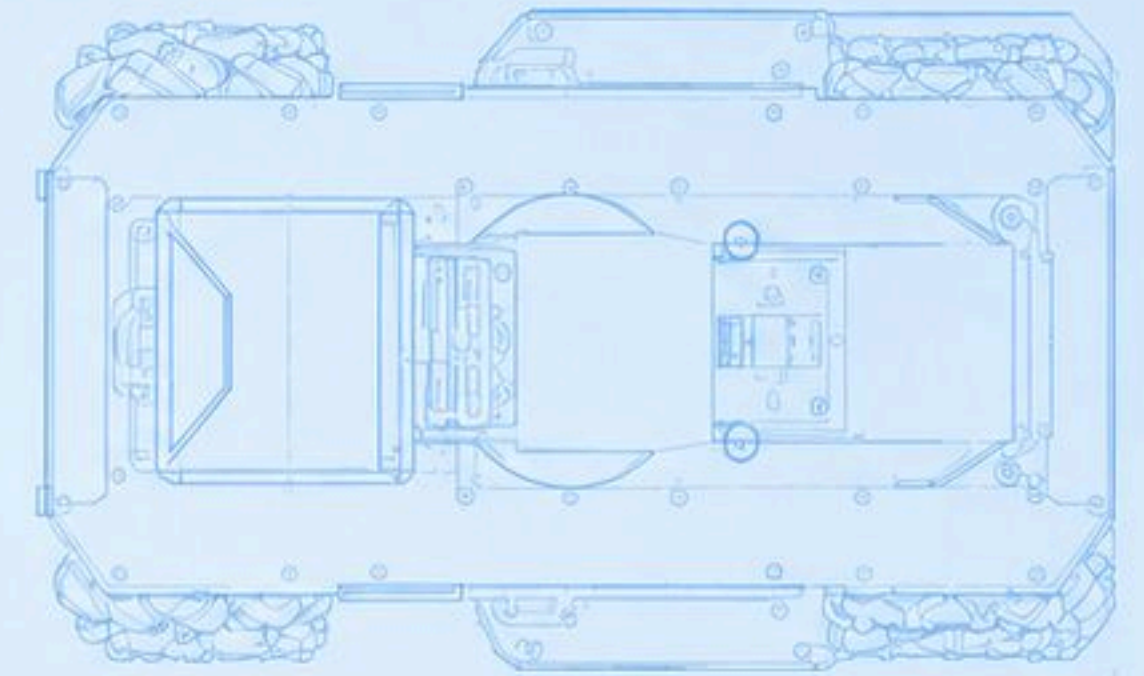
AUDIX User Manual

Autonomous Warehouse Inventory Auditing Robot

	Manual Version: 1.0
	Course: MCT332 / MCT334 — Design of Mechatronics 2
	Semester: Spring 2026
	Institution: Ain Shams University — Faculty of Engineering
	Robot Name: AUDIX
	Contact Email: support@audix.ai
	Phone: +20 10 92273762



This manual provides safety, operation, maintenance, troubleshooting, and technical information for the AUDIX warehouse inventory auditing robot.



Introduction

1. Introduction

1.1 Purpose of the Manual

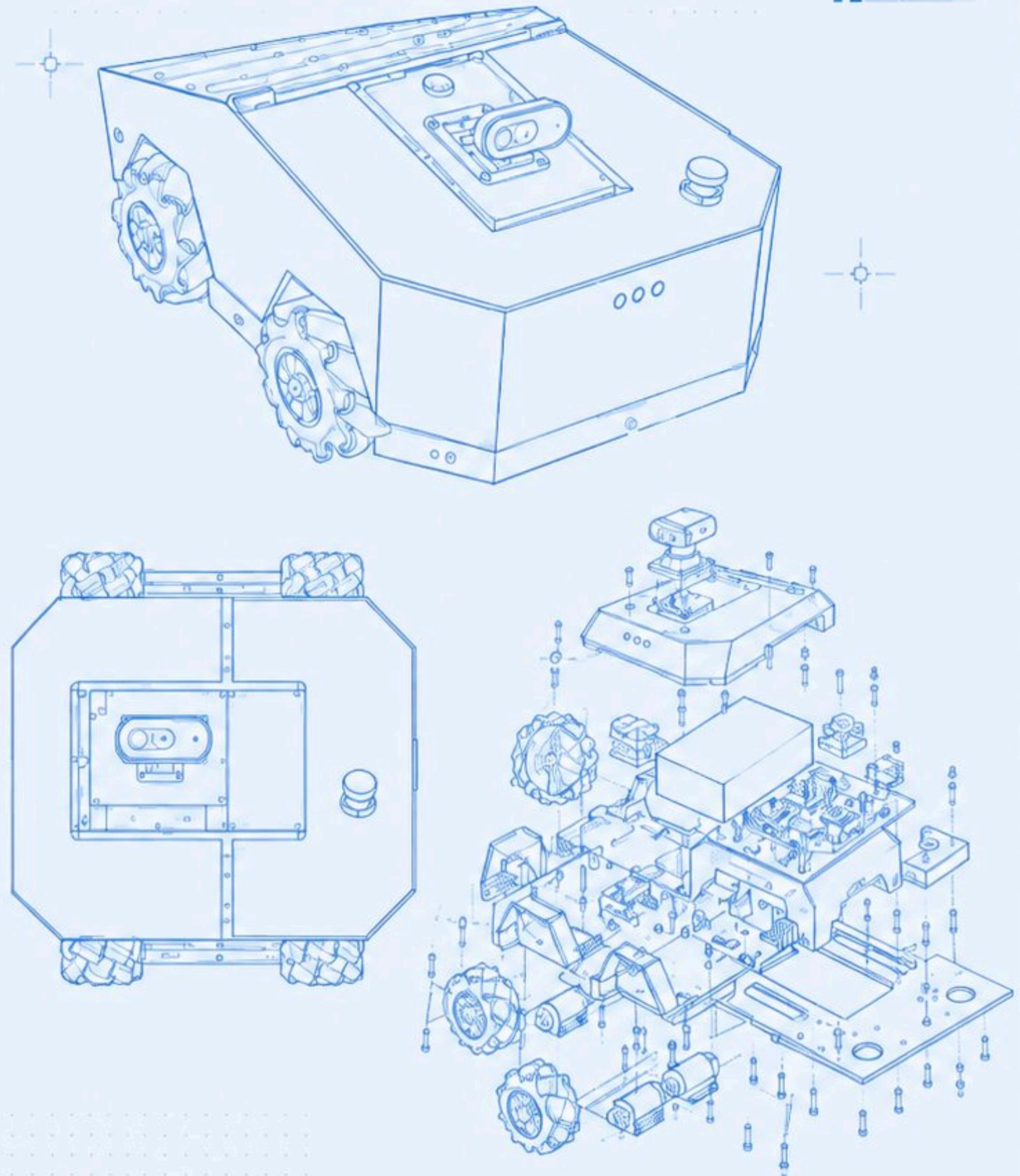
This user manual provides the information needed to safely operate, monitor, and maintain AUDIX, a semi-autonomous warehouse inventory auditing robot. It explains the robot's main functions, operating procedure, safety precautions, GUI usage, charging process, and troubleshooting guidance. The manual is intended to help users understand the system and use it correctly in a safe and reliable manner.

1.2 Intended Users

This manual is intended for operators, supervisors, maintenance personnel, and technical evaluators who interact with or assess the AUDIX robot. It is written to be clear and practical, allowing users to understand the system without needing advanced knowledge of ROS2, embedded control, or robot programming. However, safe handling of electrical and mechanical systems is still expected.

1.3 System Overview

AUDIX is designed to assist in warehouse inventory inspection by navigating through aisles, scanning shelves, and helping detect missing, misplaced, mismatched, understocked, or overstocked items. The robot combines a mobile base, vision system, sensors, Raspberry Pi high-level control, ESP32 low-level control, and Wi-Fi GUI monitoring in one integrated platform. Together, these subsystems enable organized, semi-autonomous inventory auditing with improved efficiency and reduced manual effort.



Safety Instructions

2. Safety Instructions

2.1 General Safety Precautions

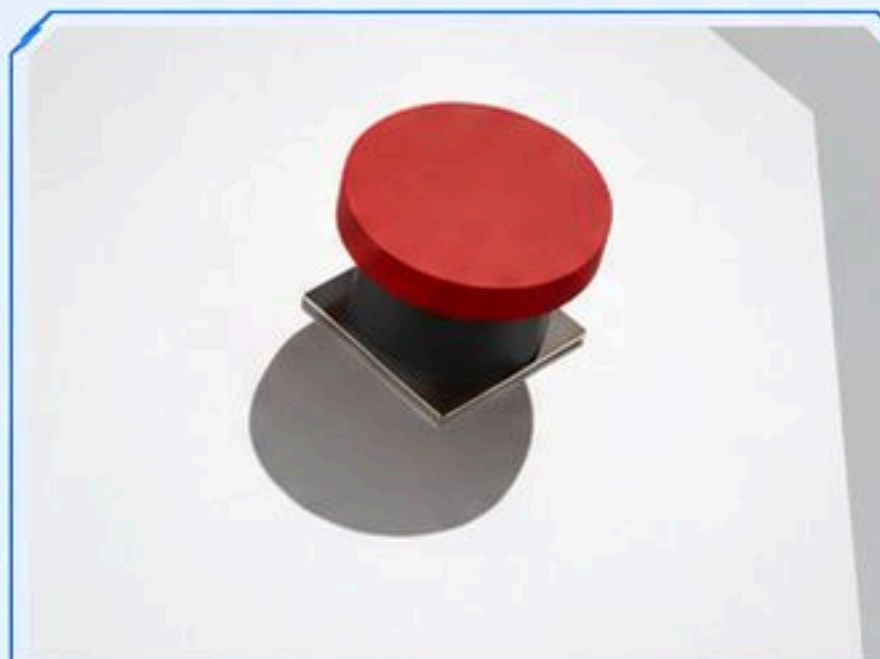
Audix should be operated only in a clear indoor environment and only after the robot has been inspected for safe mechanical and electrical condition. Operators must keep hands, loose clothing, and tools away from the wheels, moving parts, and top mechanism. The robot must be monitored during operation and must be stopped immediately if abnormal motion, unusual sound, or unsafe behavior is observed.

2.2 Emergency Stop Usage

The emergency stop button is mounted on the top surface of the robot and is used to immediately remove actuator power in unsafe situations. It must be pressed if the robot moves unexpectedly, approaches a collision, or if any mechanical or electrical hazard is observed. After activation, the cause of the fault must be checked before restarting the robot.

2.3 Battery and Electrical Safety

Audix uses a 12V Li-ion battery system. The electrical system includes an overcurrent fuse and a reverse-polarity protection circuit. The robot should not be operated if wires, connectors, or charging elements are loose, damaged, or overheated. The battery area should be inspected before operation and the robot should be powered down before maintenance.



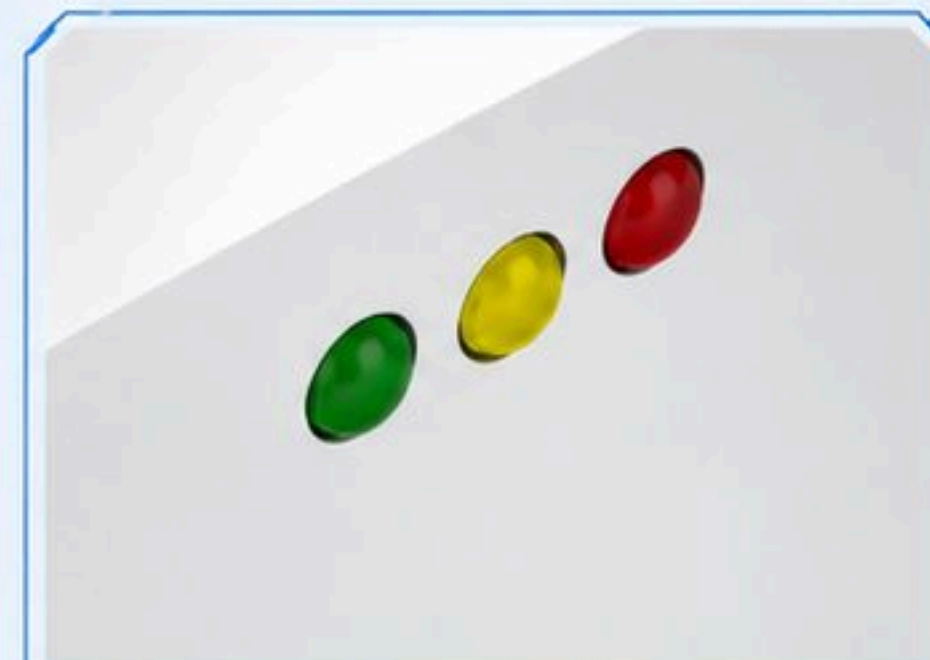
> Emergency Stop



> Moving Parts



> Battery & Electronics Access



> State Indicators



Indicators & Alerts

- Green: Ready
- Yellow: Moving
- Red: Scanning
- Buzzer: Obstacle detected



Robot Overview

3. Robot Overview

3.1 Robot Mission

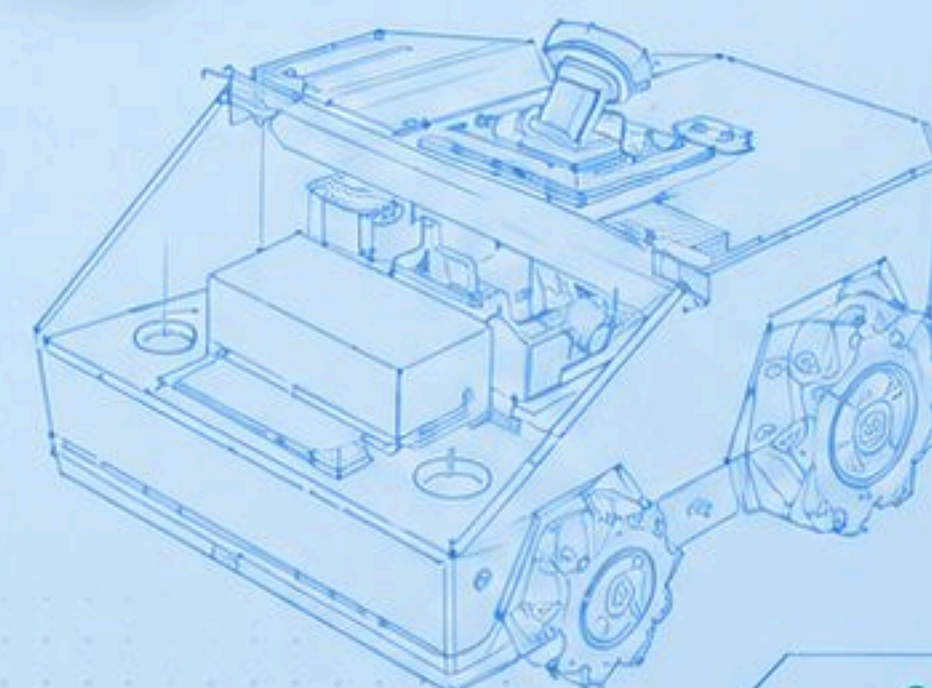
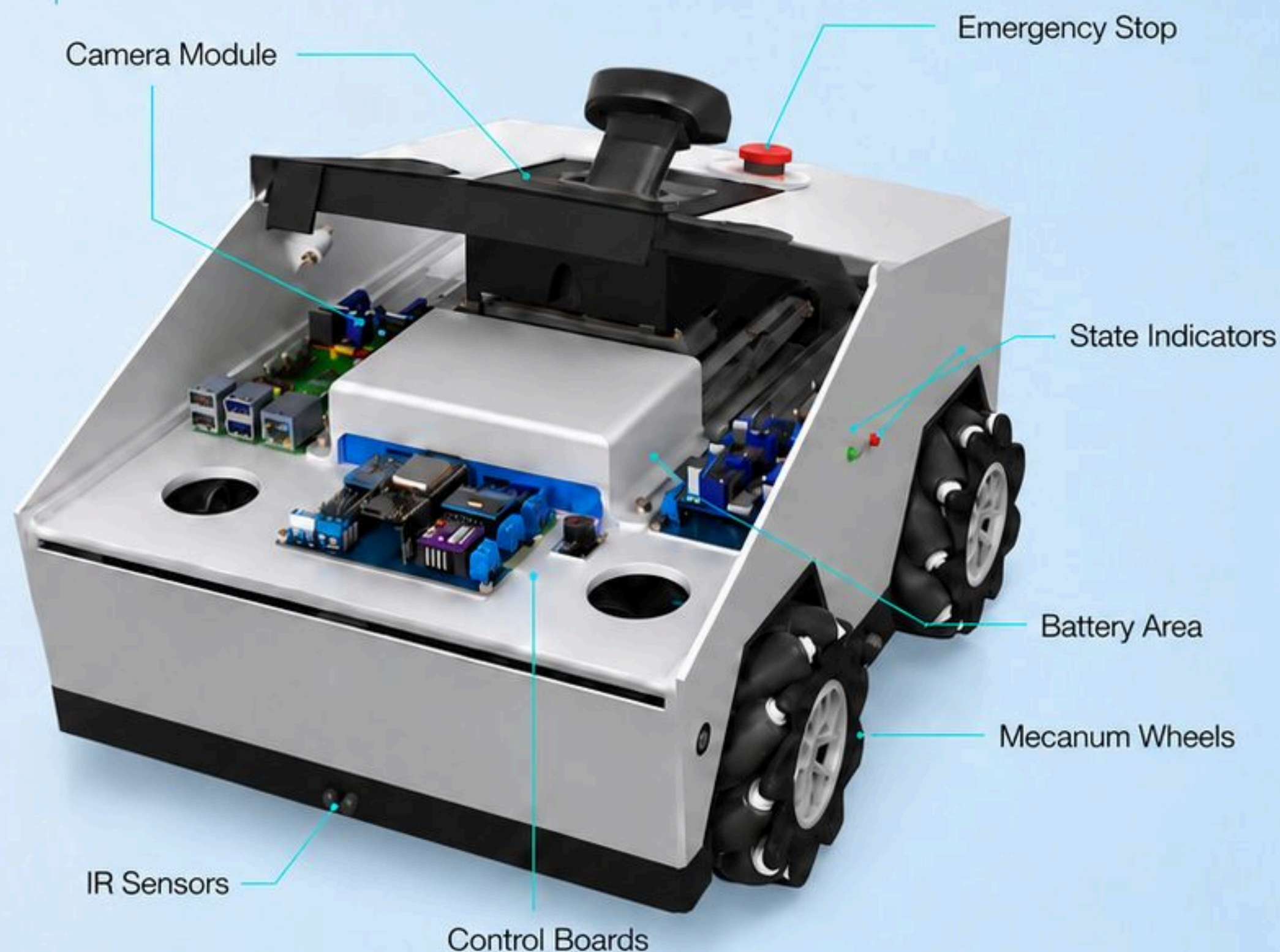
AUDIX is a semi-autonomous warehouse inventory auditing robot designed to navigate warehouse aisles, inspect shelves, verify inventory status, and report discrepancies. Its mission is to reduce manual auditing effort by combining mobility, sensing, and intelligent decision-making in one integrated platform.

3.2 Main Components

AUDIX consists of a mobile chassis, mecanum wheels, camera module, emergency stop button, state indicators, electronics compartment, battery system, Raspberry Pi, ESP32, sensors, and motor drivers. These components work together to support navigation, inspection, control, and mission execution.

3.3 System Overview

The robot integrates mechanical motion, electrical power distribution, embedded control, and high-level software coordination. Raspberry Pi and ROS2 manage the mission logic, while ESP32 handles low-level control. Together with the vision system and Wi-Fi GUI monitoring, AUDIX performs organized and efficient warehouse auditing.



Mechanical Overview

3.3 Mechanical System Overview

3.3.1 Chassis & Structure

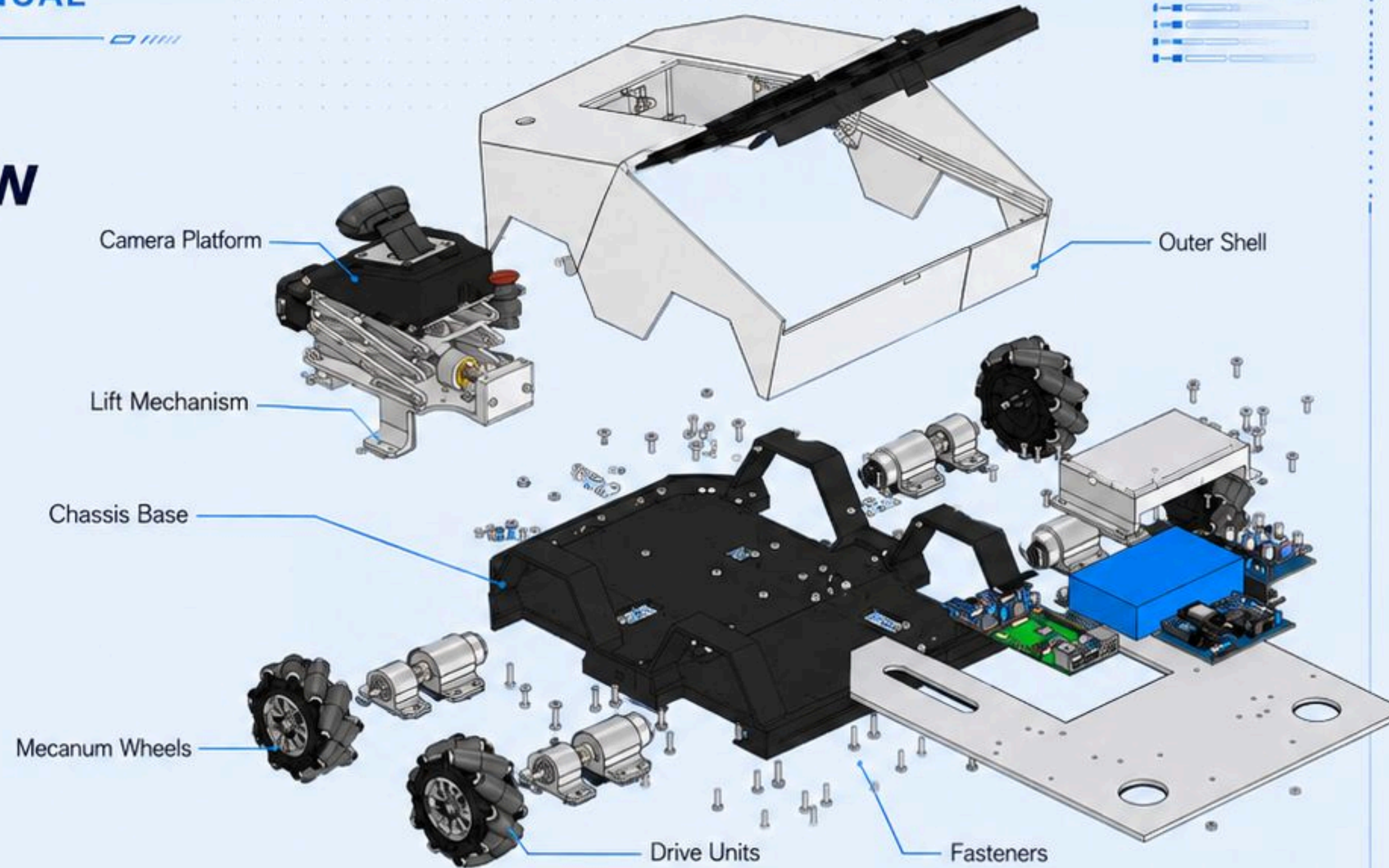
AUDIX uses a compact mobile chassis designed for warehouse aisle navigation. The robot features an enclosed outer body with an accessible upper cover, a stable base structure, and integrated mounting points for all mechanical and electronic subsystems, ensuring durability, protection, and easy maintenance.

3.3.2 Wheel Configuration & Drive Layout

The robot uses mecanum wheels to provide omnidirectional maneuverability, enabling smooth movement and precise positioning in confined warehouse spaces. The distributed wheel and drive layout, as shown in the supporting views, ensures balanced traction, stability, and efficient power distribution across the platform.

3.3.3 Active Inspection Mechanism

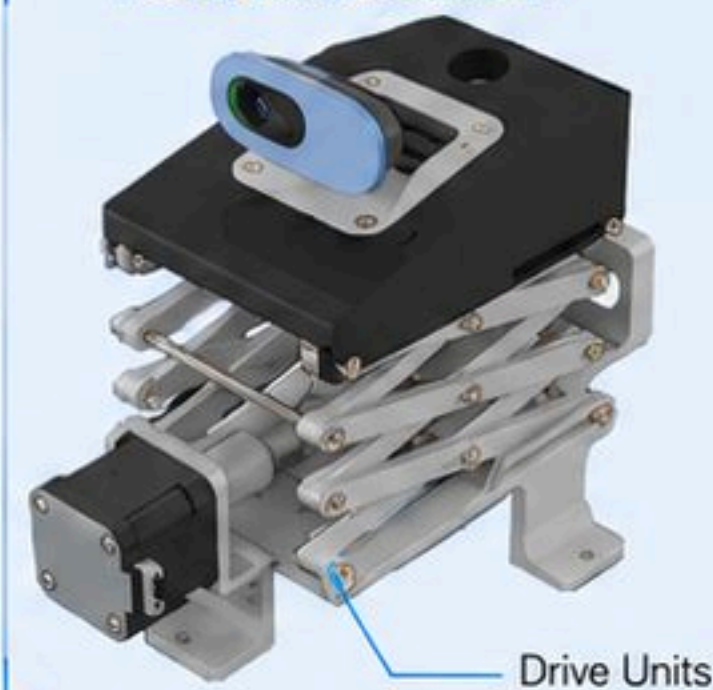
AUDIX includes an actively controlled lift mechanism that directly supports the mission by adjusting the camera position for shelf inspection. The mechanism is fully integrated into the mechanical structure and is designed for compact motion, reliable actuation, and stable support of the camera platform.



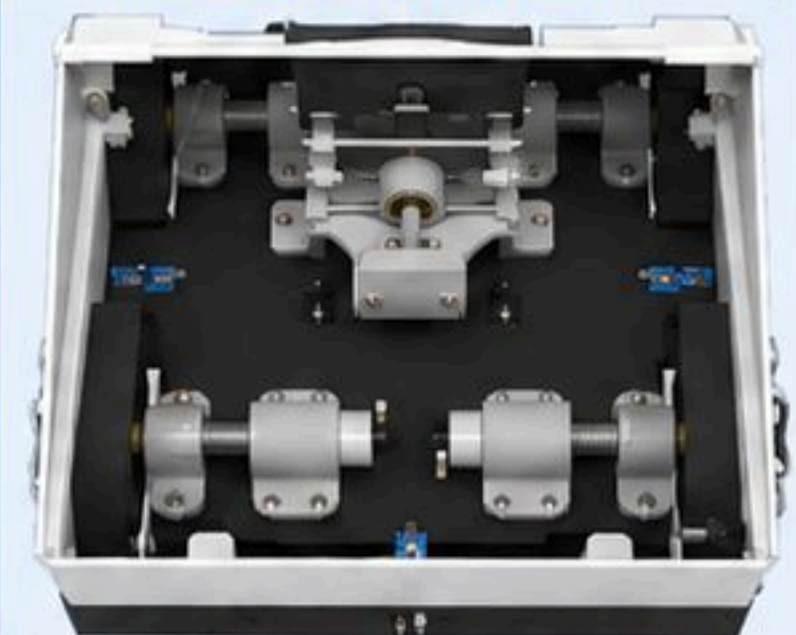
Drive Layout (Top View)



Active Lift Mechanism



Internal Layout (Top View)



Electrical & Software Overview

3.4 Electrical System Overview



3.4.1 Custom PCB & JST Wiring

AUDIX uses a custom designed PCB instead of breadboards to ensure reliable electrical performance. All connections between subsystems are made using JST connectors for clean wiring, easier maintenance, secure connections, and fast subsystem inspection or replacement.



3.4.2 Power & Safety

The robot is powered by a 12V Li-ion battery. A fuse provides overcurrent protection, and a reverse polarity protection circuit safeguards the electronics from incorrect connections. The emergency stop physically removes actuator power only, ensuring the system can be safely disabled during critical situations. Cable organization and connectorized wiring simplify maintenance and allow for easy verification of connections.



3.4.3 Indicators & Charging

Status indicators provide clear operational feedback: Green = Ready, Yellow = Moving, Red = Scanning. The buzzer activates when an obstacle is detected to alert nearby personnel. An operational charging port is provided on the chassis side for convenient and safe recharging of the battery.

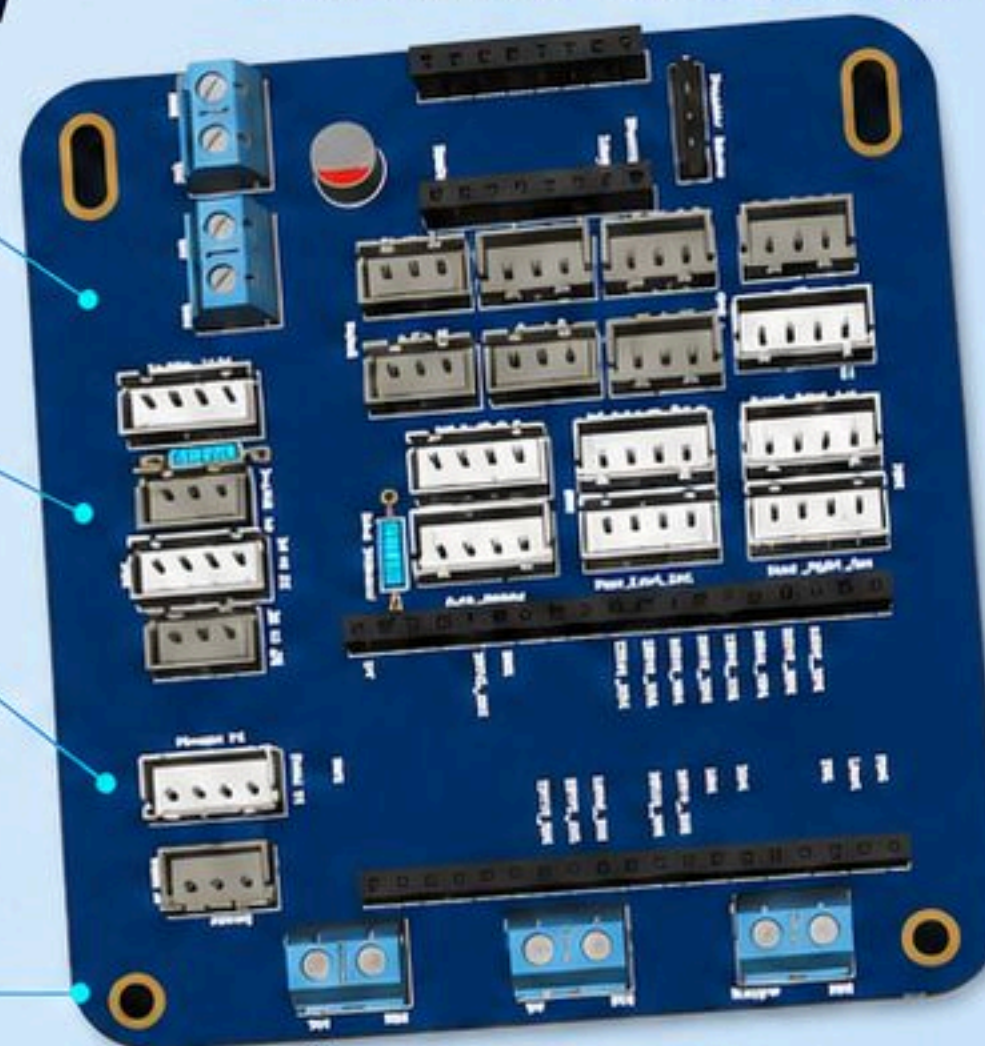
JST Connectors
Clean, secure wiring for all subsystems

Power Input
12V power input with fuse protection

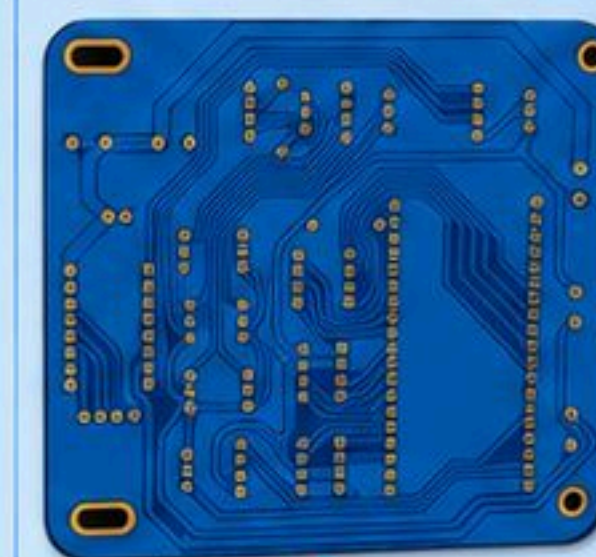
Motor / Sensor Ports
Dedicated ports for motors, sensors, and actuators

Mounting Points
Built-in mounting holes for stable installation

Custom Control PCB – Top / Component Side



PCB – Bottom / Trace Side



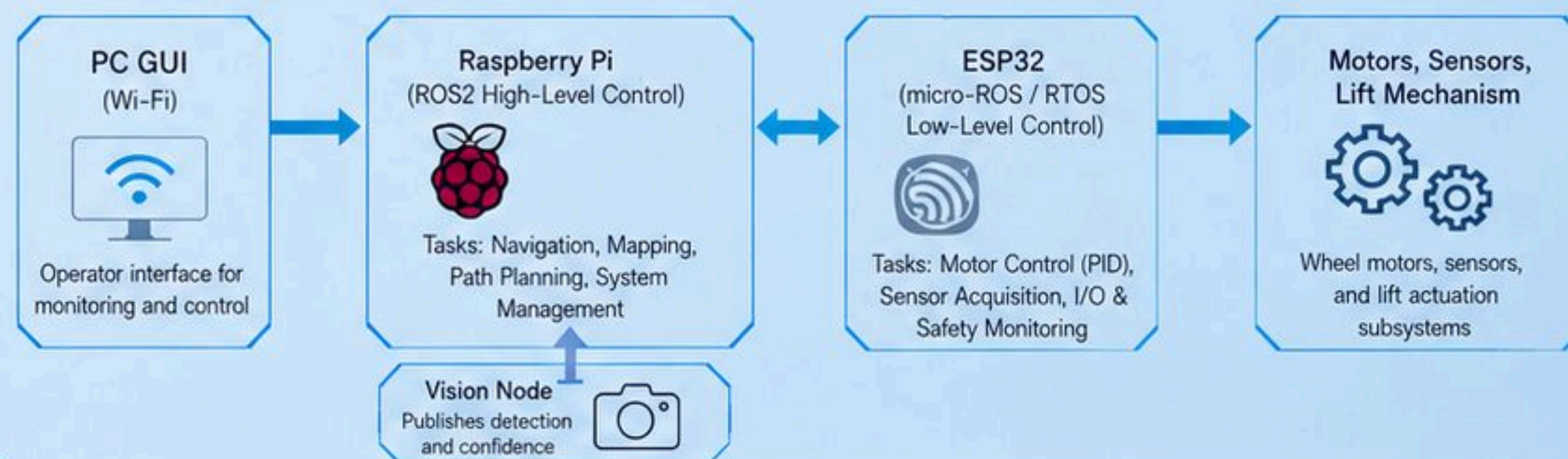
Signal Routing
Optimized trace layout for signal integrity and reliability

Charging Port (Side View)



Charging Port
Operational charging port located on the side of the chassis

3.5 Software System Overview



AUDIX integrates a well-organized electrical architecture with robust software coordination. High-level decision making and navigation are handled by ROS2 on the Raspberry Pi, while the ESP32 manages real-time motor control, sensor acquisition, and safety-critical functions using micro-ROS and RTOS. Together, this layered approach ensures reliable communication, precise control, and safe operation in dynamic warehouse environments.

Pre-Operation Checklist

Complete the following checks before each mission to ensure safe and reliable operation.



4.1 Mechanical Inspection

Inspect the chassis, mecanum wheels, lift module, and fasteners for visible damage, looseness, or obstruction before operation.



4.2 Electrical Inspection

Check the power input, wiring integrity, and emergency stop accessibility, and confirm that all electrical connections are secure.



4.3 Battery Check

Confirm that the 12V Li-ion battery is sufficiently charged and properly connected before starting the mission.



4.4 Sensor Check

Ensure that the camera and IR sensors are clean, unobstructed, and responding correctly before deployment.



4.5 GUI and Wi-Fi Connection Check

Verify that the GUI is connected through Wi-Fi, live robot data is updating, and the selected route and operating mode are correct.

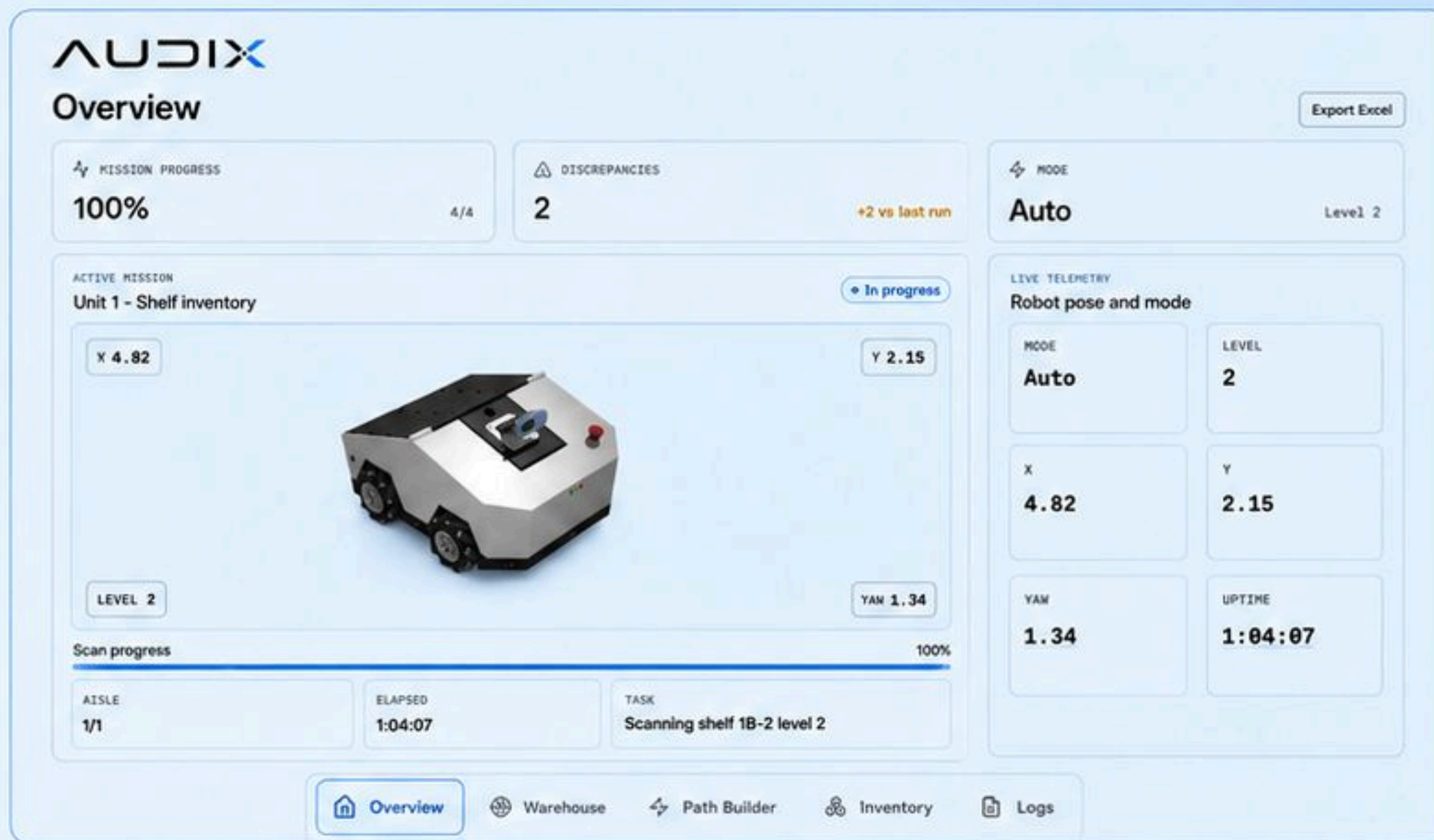


FIGURE 4.1 Overview Dashboard

Monitor mission status, robot telemetry, and system health.



FIGURE 4.2 Path Builder

Review the selected route and shelf targets.

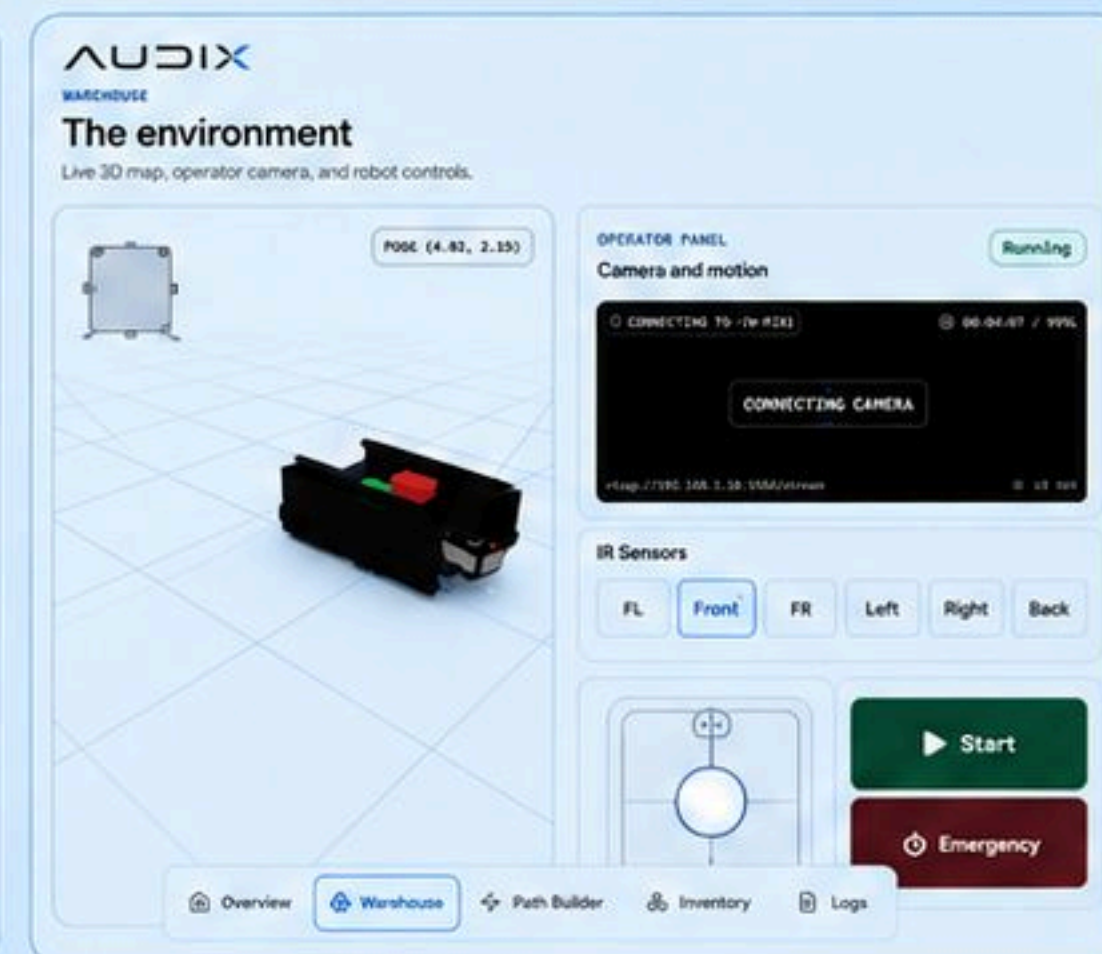


FIGURE 4.3 Warehouse / Operator Panel

Confirm camera/IR sensor status and control readiness.

5. OPERATING INSTRUCTIONS

Follow the steps below to power ON AUDIX, connect the GUI, select the mode or goal, and start the mission.

5.1 POWERING ON THE ROBOT

Turn ON the main ON/OFF switch located on the robot. Wait for the Raspberry Pi and ESP32 to initialize. The state indicators will show Power-Good (green). Once initialization is complete, the robot is ready for GUI connection.



5.2 CONNECTING THE GUI

Ensure your PC is connected to the same Wi-Fi network as AUDIX. Launch the AUDIX GUI. When connected, the [Overview](#) screen will display live robot data, mission progress, and system status. Verify that Mode, Pose (X,Y,Yaw), and Mission Progress are updating.



5.3 SELECTING THE OPERATION MODE

Choose the required operation mode:

- **Auto Mode** – The robot autonomously navigates and executes the mission.
- **Manual Mode** – The robot is controlled manually using the GUI.

Select the mode from the GUI before proceeding.

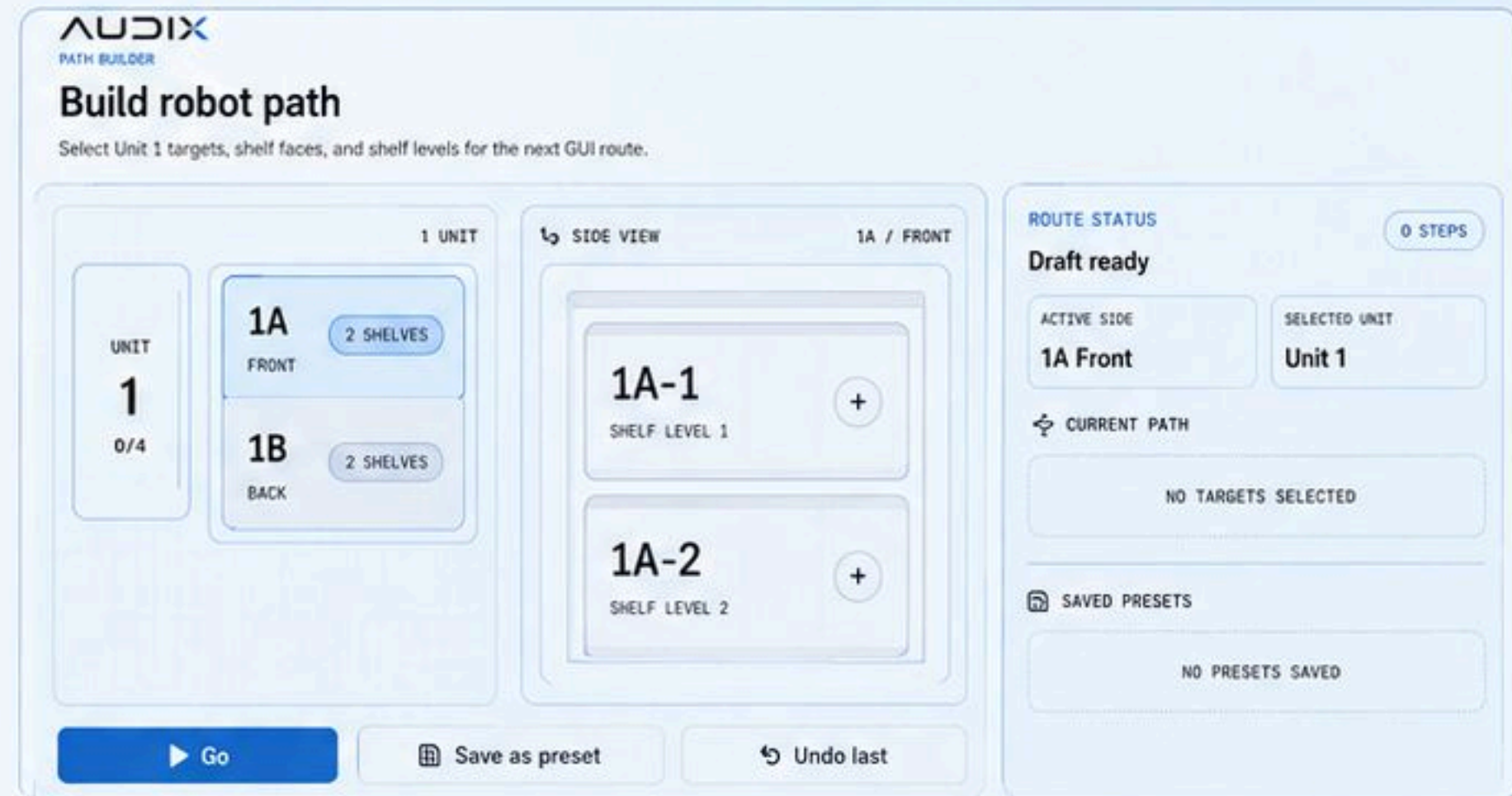


5.4 STARTING THE MISSION

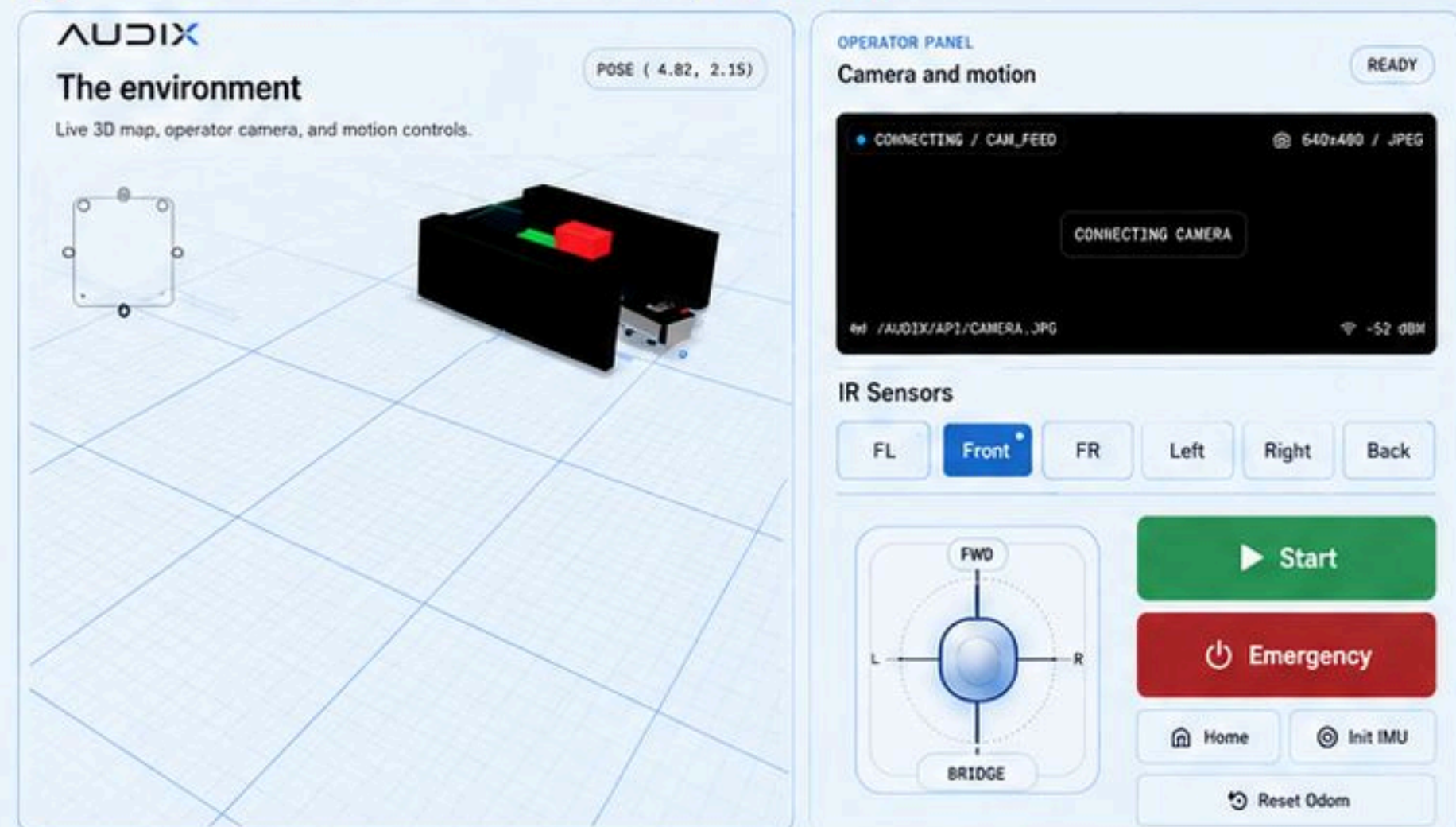
For Auto Mode: Use the [Path Builder](#) to select the target unit, shelf face, and shelf level. The selected route will be displayed in the [Route Status](#) panel. Click [Go](#) to activate the route. Then, go to the [Warehouse](#) page and press [Start](#) to begin mission execution.

For Manual Mode: Go to the [Warehouse](#) page and use the controls to operate the robot.

1 SELECT MISSION TARGET (Path Builder)



2 START MISSION / MANUAL CONTROL (Warehouse)



SAFETY REMINDERS



Emergency Stop

The physical E-Stop button cuts actuator power immediately.



State Indicators

Confirm [Power-Good](#) (green) and system status before operation.



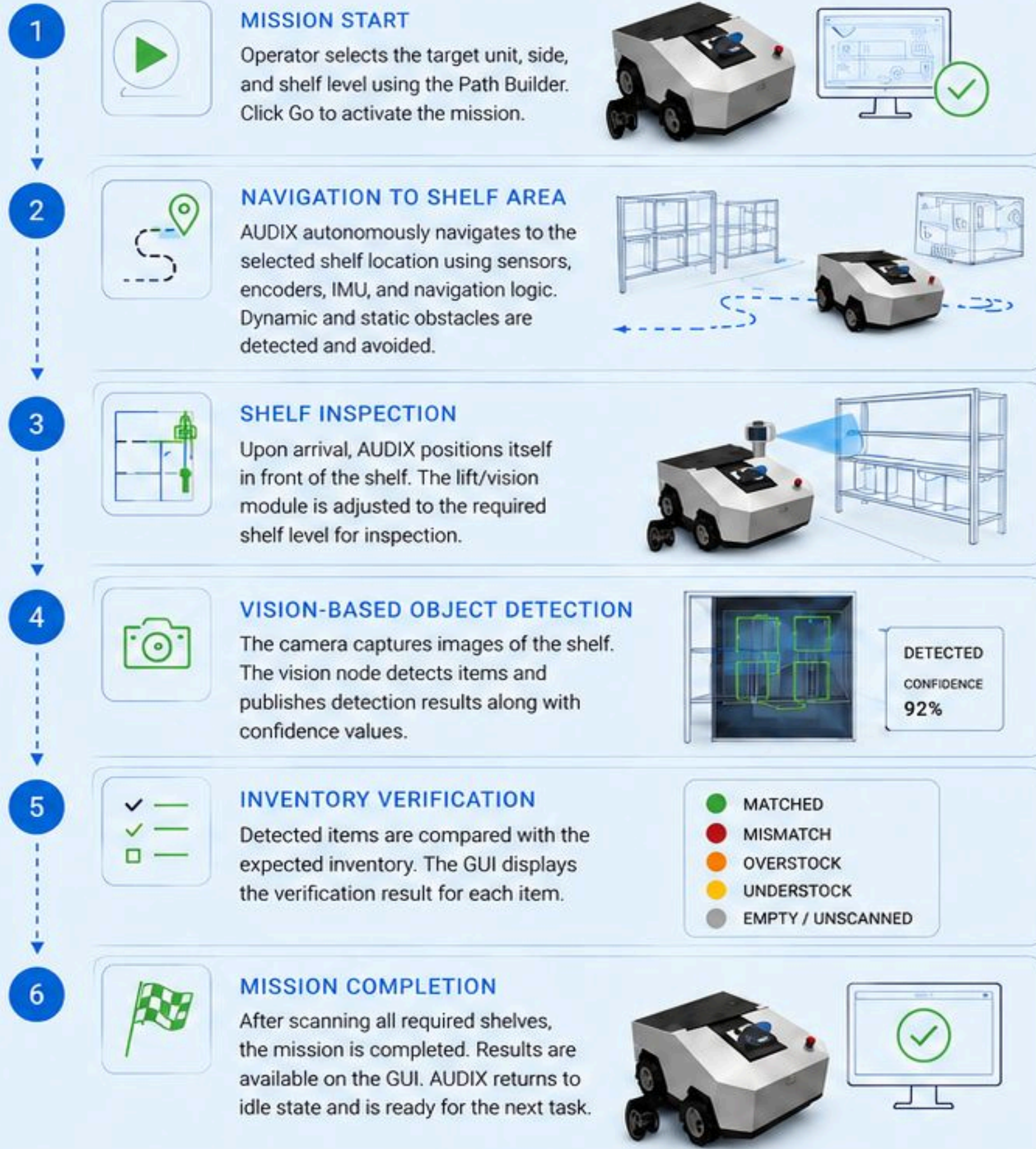
Power Stability

Ensure battery is charged and connections are secure.



7. ROBOT MISSION WORKFLOW

This section describes the complete mission sequence of AUDIX from start to finish.



AUTONOMOUS OPERATION

During the mission, AUDIX makes decisions autonomously. Operator input is only required to start/stop the mission, select goals, or press Emergency Stop.



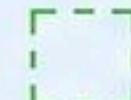
SAFETY FIRST

AUDIX continuously monitors its environment and system health. If any unsafe condition occurs, the robot will halt and alert the operator.



Wi-Fi Connection

GUI communicates with AUDIX over Wi-Fi network.



Vision Module

Directly supports the mission through detection and analysis.

The GUI provides all necessary tools to execute and monitor the mission.

1 SELECT TARGET (Path Builder)

AUDIX Build robot path
Select unit, target, shelf level, and start level for the new GUI path.

UNIT: 1
SHELF: 1A
SHELF LEVEL: 14-1
SHELF LEVEL: 14-2

ROUTE STATUS: Path ready
START SIDE: 1A Front
SELECTED UNIT: Unit 1
CURRENT PATH: NO TARGETS SELECTED
SAVED PRESETS: NO PRESETS SAVED

Go Save as preset Undo last

Choose the unit, side (front/back), and shelf level. The selected route and mission status are shown in the Route Status panel.

2 NAVIGATION & MONITORING (Warehouse)

AUDIX The environment
Live 3D map, waypoint routes, and robot controls.

POSE (x, y, z, h, h, h, h)

CAMERA & ENV VISION: Cameras are ON VISION
1 CAMERA: FRONT (FEED)
3 DEPTH + (LIGHT)
CONNECTING CAMERAS
AI - ARCHITECTURAL_3D
0° - 0° / 180°

IR SENSORS: Front, FR, Left, Right, Back
Start Emergency
SPEED: 100% 01:25:00

Monitor the robot in real time. View live camera feed, IR sensor status, robot pose, and environmental map. Use Start to begin and Emergency to stop immediately.

3 INVENTORY VERIFICATION (Inventory)

AUDIX INVENTORY
Review scanned results and verification status.

1A-1-1: MATCHED Confidence: 95%
1A-1-2: MISMATCH Confidence: 72%
1A-1-3: OVERSTOCK Confidence: 85%
1A-1-4: UNDERSTOCK Confidence: 81%
1A-1-5: EMPTY Confidence: ---

Each shelf location shows the verification result and confidence percentage for the detected items. Results are based on your expected inventory.

4 MISSION LOGS & SUMMARY (Logs)

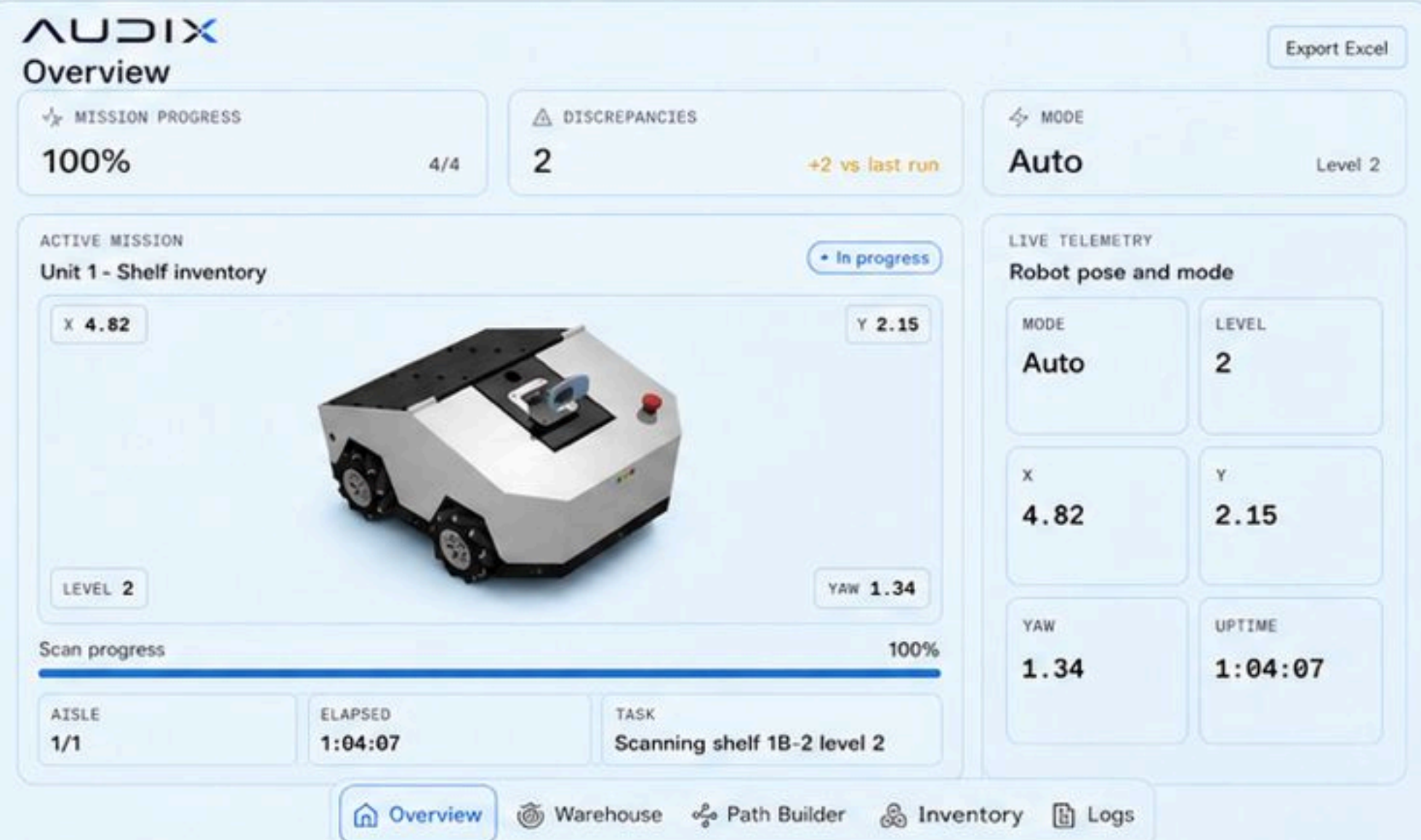
AUDIX Logs
Mission events, discoveries, and system messages

Time	LEVEL	LOCATION	MESSAGE	CONFIDENCE	ACTION
10:12:15	1A-1-1	1A-1-1	Matched	100%	
10:12:18	Mismatch	1A-1-2	Below threshold	72%	Review
10:12:19	Overstock	1A-1-3	Exceeds Expected	85%	Review
10:12:19	Understock	1A-1-4	Running Low	81%	Review
10:12:22	Info	1A-1-5	No item Detected	---	Review

View detailed mission logs, discrepancy types, confidence, and available actions. Click Review to re-check a location if needed.

5.5 MONITORING THE MISSION (Auto Mode)

Once the mission is started, monitor the robot using the Overview screen. This screen provides real-time information about mission progress, robot pose, mode, telemetry, and system status.



What to Monitor

- ✓ **Mission Progress** – Percentage and task status
- ✓ **Robot Pose (X, Y, Yaw)** – Position and orientation
- ✓ **Mode & Level** – Current operation mode and autonomy level
- ✓ **Discrepancies** – System alerts or issues
- ✓ **Uptime** – Total mission time
- ✓ **Telemetry** – Live system data

5.7 POWERING OFF THE ROBOT

After the mission is stopped and the robot is is safe OFF the robot.

- 1 Ensure the robot is fully stopped.
- 2 Close the GUI application.
- 3 Turn OFF the ON/OFF switch.

ON / OFF SWITCH



- 4 Disconnect the battery if required.



Important

Always follow the proper shutdown procedure to protect the system and ensure safety.

5.6 STOPPING THE MISSION

Stop the mission only when the task is complete or if a safe stop is required.

A Normal Stop (Through GUI)

In Auto Mode: Mission will complete automatically and can be stopped using the GUI.

In Manual Mode: Press the Stop button if the mission is running.



Stop Mission

Mission will stop safely and the robot will halt.

B Emergency Stop (Physical E-Stop)

In any unsafe situation, press the physical E-Stop button immediately.

This will cut actuator power and halt all robot motion.



E-STOP

Cuts actuator power immediately. Use only in emergencies.

C Emergency Button (GUI – Manual Mode)

In Manual Mode, you can also press the Emergency button on the GUI. This will stop the robot safely.



Emergency

Stops the robot safely (software stop).



Note: After any type of stop, wait until the robot is completely stationary before performing the next operation.

8. CHARGING PROCEDURE

Follow the steps below to safely charge AUDIX. Always use the provided charger and follow safety guidelines.

- 1 STOP**
STOP THE MISSION
 Ensure the robot is in idle state. Stop all operations from the GUI.
- 2**

POWER OFF AUDIX
 Press the E-Stop button or power switch to cut off power to all actuators before charging.
- 3**

INSPECT THE CHARGING PORT
 Check the charging port and cable for any damage, dirt, or moisture before connecting.
- 4**

CONNECT THE CHARGER
 Align the charger connector with the charging port and insert completely. Ensure a secure connection.
- 5**

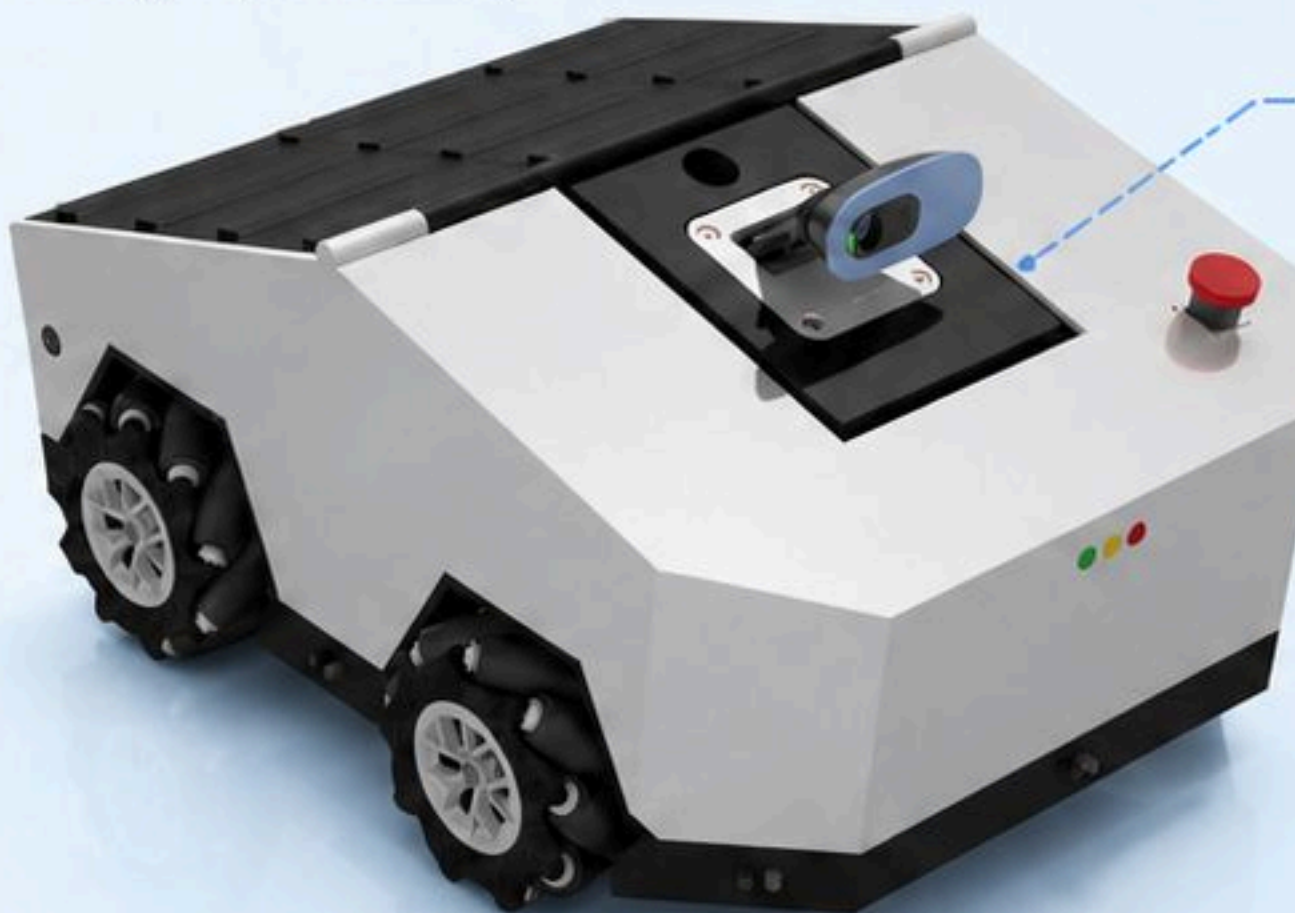
MONITOR CHARGING
 The status LEDs will indicate charging progress. Charging time is approx. 2–3 hours for full charge.
- 6**

DISCONNECT AFTER CHARGING
 Once charging is complete (green LED ON), unplug the charger from the port first, then from the power outlet.
- 7**

CONFIRM READY STATE
 Verify power level on GUI and ensure AUDIX is ready for the next mission. Store the charger safely.

8.1 CHARGING PORT LOCATION

The charging port is located on the top panel of AUDIX. Use only the provided charger.



 Ensure the protective cap is closed when the port is not in use.

8.2 CHARGING SYSTEM OVERVIEW



8.3 CHARGING SAFETY NOTES

-  Do not charge while the robot is operating.
-  Do not charge in wet or humid environments.
-  Do not use damaged cables or connectors.
-  Do not force the connector into the port.
-  Use only the provided charger and power outlet.
-  Ensure ambient temperature is 0°C – 40°C.
-  Disconnect the charger after charging is complete.

 Proper charging ensures battery health, system safety, and reliable mission performance.

GUI User Guide

Understanding the main interface screens, displayed robot data, mission controls, manual mode, and status feedback.

1 6.1 GUI Main Screen

The Overview screen is the main dashboard for checking mission progress, mode, discrepancies, robot pose, scan progress, and live telemetry.

2 6.2 Displayed Robot Data

The Overview screen displays mission progress, robot mode, robot pose (X, Y, Yaw), level, uptime, discrepancies, active mission, and scan progress.

3 6.3 Mission Controls

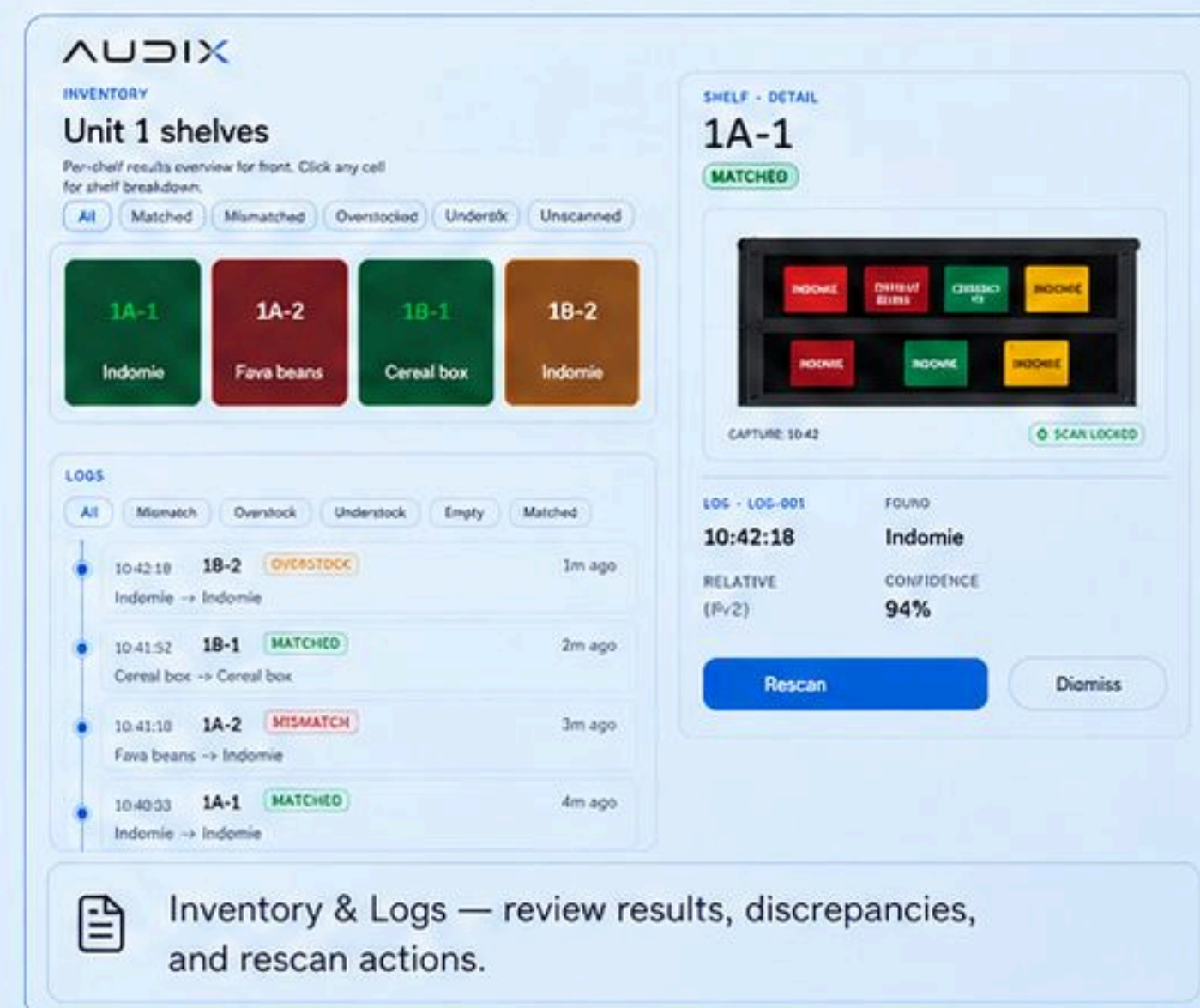
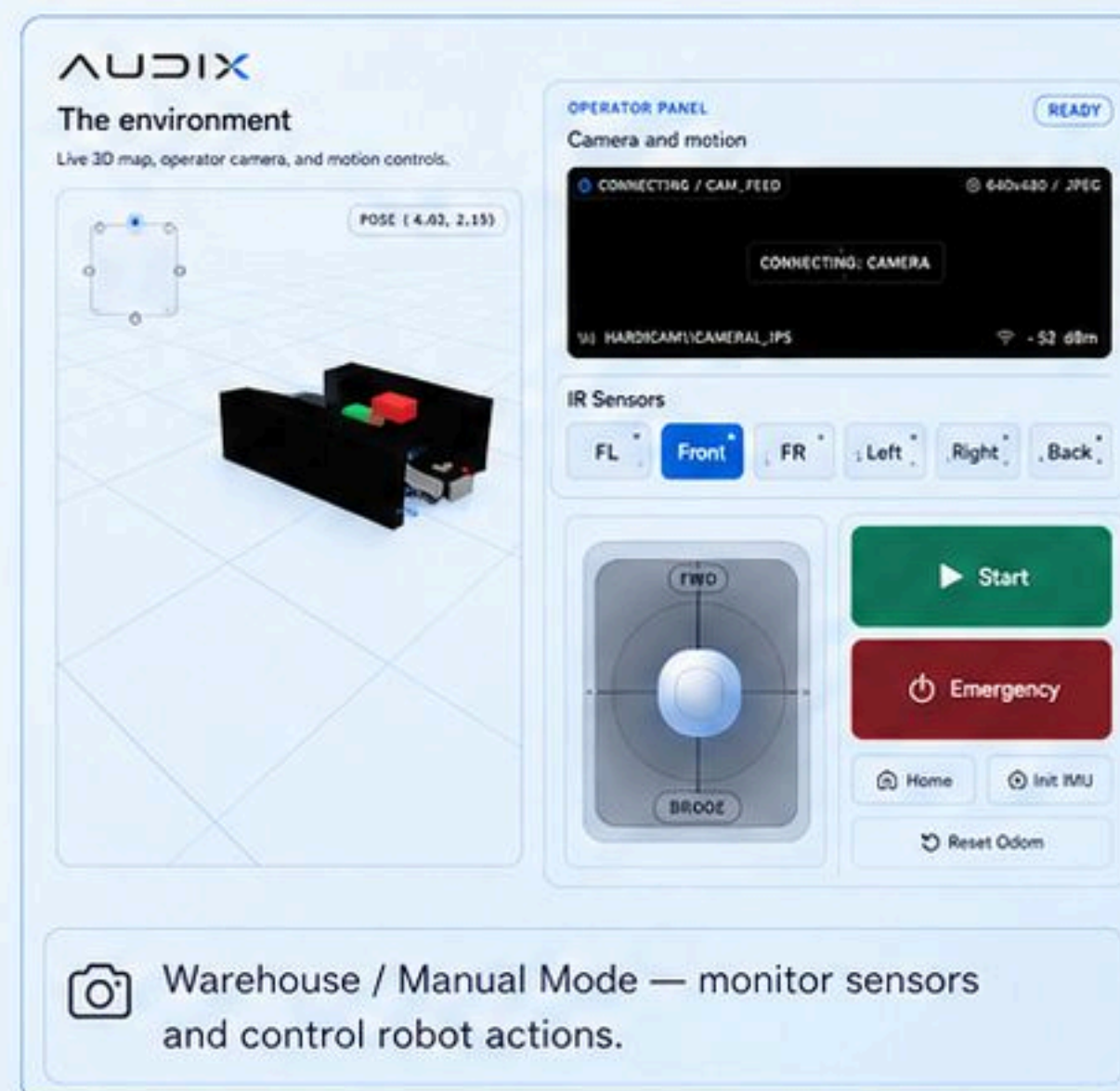
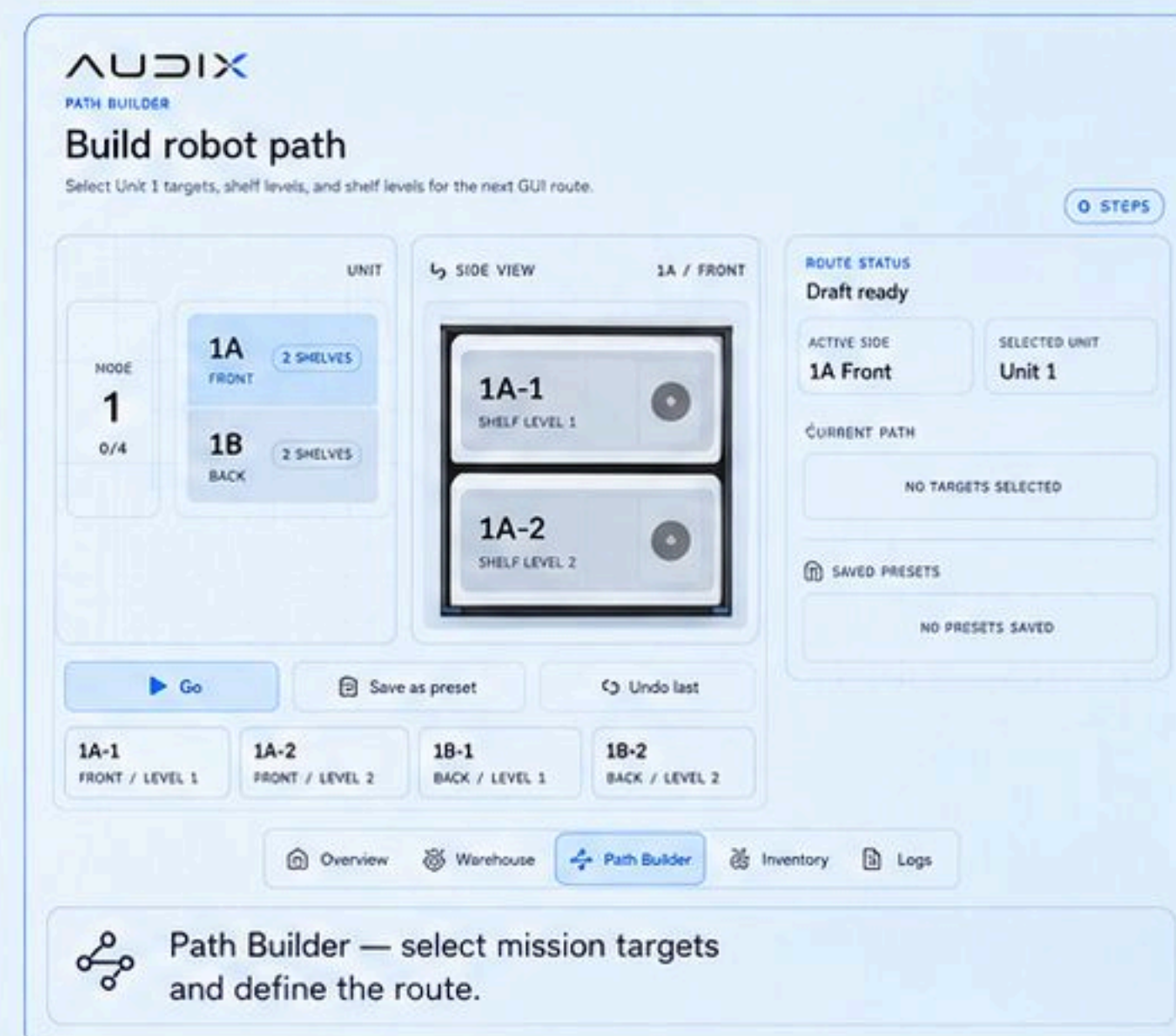
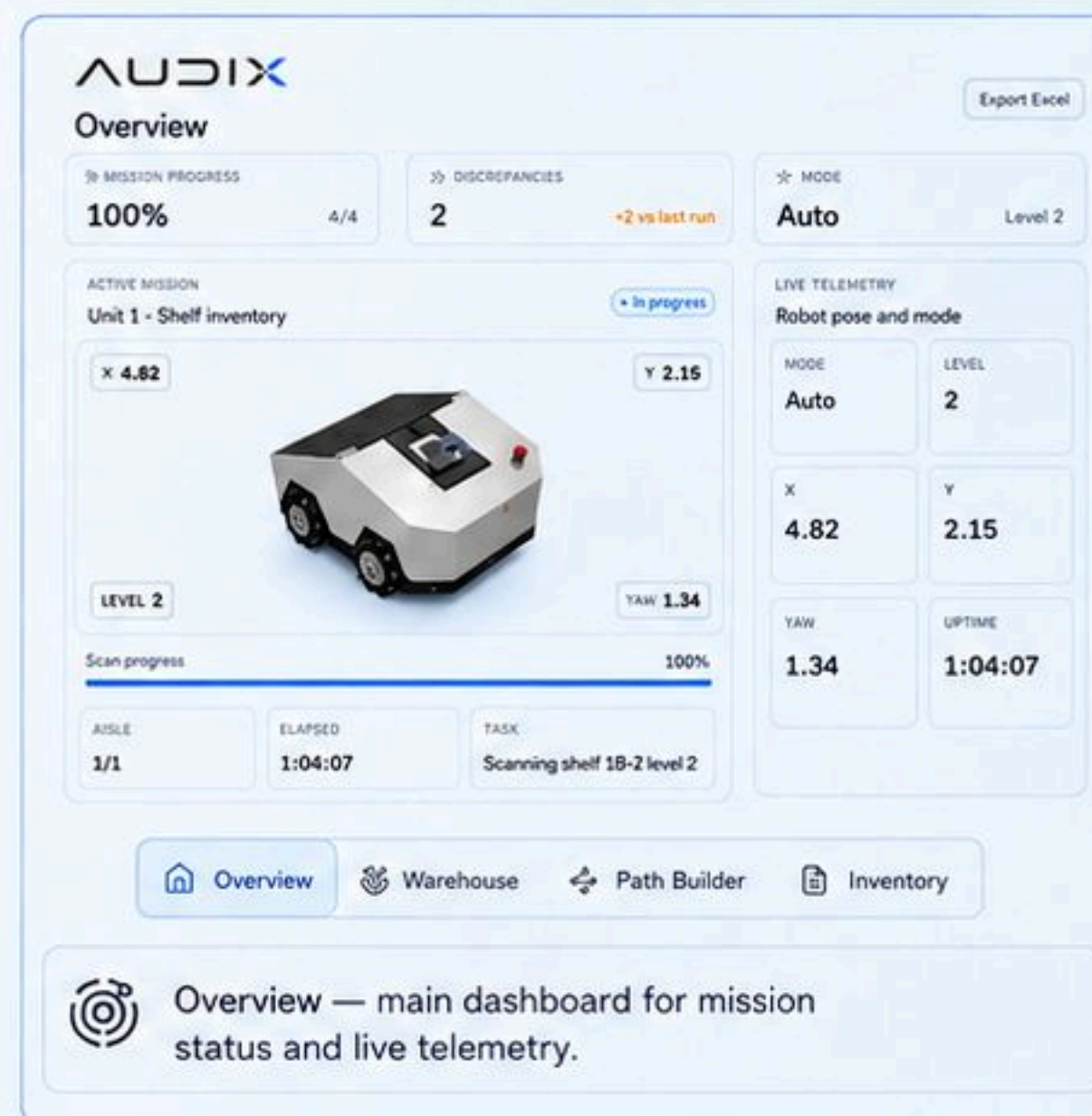
The Path Builder screen is used to select the unit, side, and shelf level, build the route, and start the planned mission path.

4 6.4 Manual Mode

The Warehouse screen provides manual operation tools including camera feed, IR sensor status, motion controls, Start, Emergency, Home, Init IMU, and Reset Odom.

5 6.5 Error and Status Messages

Inventory and Logs screens help the operator review shelf results, confidence, matched or mismatch status, overstock and understock conditions, scan results, and rescan actions.



10. TROUBLESHOOTING

Use the table below to identify common issues and follow the recommended actions.
If the problem persists, contact support.

#	PROBLEM	POSSIBLE CAUSE(S)	CORRECTIVE ACTION	
1	 Robot does not power ON	- Battery is depleted. - Main power switch is OFF. - Fuse is blown. - Loose or disconnected power cable. - Reverse polarity connection.	- Charge the battery. - Toggle the main power switch ON. - Check the fuse and replace if necessary (ATO 15A). - Ensure all connections are secure. - Verify correct polarity at the charging port.	See: 6
2	 Robot does not move	- E-Stop is activated. - Manual mode is not selected. - Wheel/motor connections are loose. - Motor driver fault. - Obstacle detected or safety stop.	- Release the E-Stop (rotate clockwise). - Select MANUAL mode in the GUI. - Check motor and wheel connections. - Verify motor driver indicators. - Clear the path and retry.	See: 2
3	 GUI does not connect	- Robot and PC are not on same Wi-Fi. - Robot is OFF or not fully booted. - Incorrect IP address or port. - Firewall blocking connection.	- Connect PC to the robot Wi-Fi network. - Ensure robot is powered ON and LEDs are steady. - Verify IP address and port (default: 192.168.4.1:8888). - Allow application through firewall.	See: 7
4	 Vision detection does not work	- Camera not detected or disconnected. - Poor lighting conditions. - Vision mode not running. - Obstructed or dirty camera lens.	- Check camera connection and restart the robot. - Ensure adequate lighting. - Start/Restart the vision mode in ROS2. - Clean the camera lens gently with a soft cloth.	See: 7
5	 Robot stops during mission	- Obstacle detected by sensors. - Low battery. - Overcurrent / Overtemperature fault. - Communication lost.	- Check the environment and clear obstacles. - Check battery level in the GUI and recharge if needed. - Allow system to cool down and restart. - Verify Wi-Fi connection and restart mission.	See: 5
6	 Emergency Stop is activated	- E-Stop button was pressed. - Safety condition was triggered. - Manual activation from GUI.	- Inspect the robot and environment. - Rotate the E-Stop button clockwise to release. - Press "RESET" in the GUI if shown. - Restart the mission only when safe.	See: 2

REFERENCE: SYSTEM INDICATORS



Green : Ready
Yellow : Moving
Red : Scanning
 Buzzer : Obstacle detected

REFERENCE: E-STOP BUTTON



Rotate clockwise to release the E-Stop button.

[See Page 2](#)

REFERENCE: GUI STATUS EXAMPLES

CONNECTED	DISCONNECTED	EMERGENCY STOP
Status: CONNECTED Mode: MANUAL Battery: 84%	Status: DISCONNECTED Mode: --- Battery: ---	Status: E-STOP ACTIVE Mode: --- Battery: 78%
Normal connection	Connection lost	E-Stop is active

[See Page 7](#)

REFERENCE: LOGS (ROS2 TERMINAL)

```
[INFO] [nav2]: Navigation started
[WARN] [sensors]: Obstacle detected at 0.65 m
[ERROR] [motor_driver]: Overcurrent detected on left motor
[INFO] [system]: Emergency stop activated
[INFO] [system]: E-Stop released
[INFO] [system]: Mission resumed
[INFO] [vision]: Object detected - confidence: 0.87
```

[See Page 7](#)



IMPORTANT:

Always power OFF the robot before performing any inspection or maintenance.



Power OFF before inspection.



Do not open the chassis unless authorized.



Keep the robot away from water and liquids.

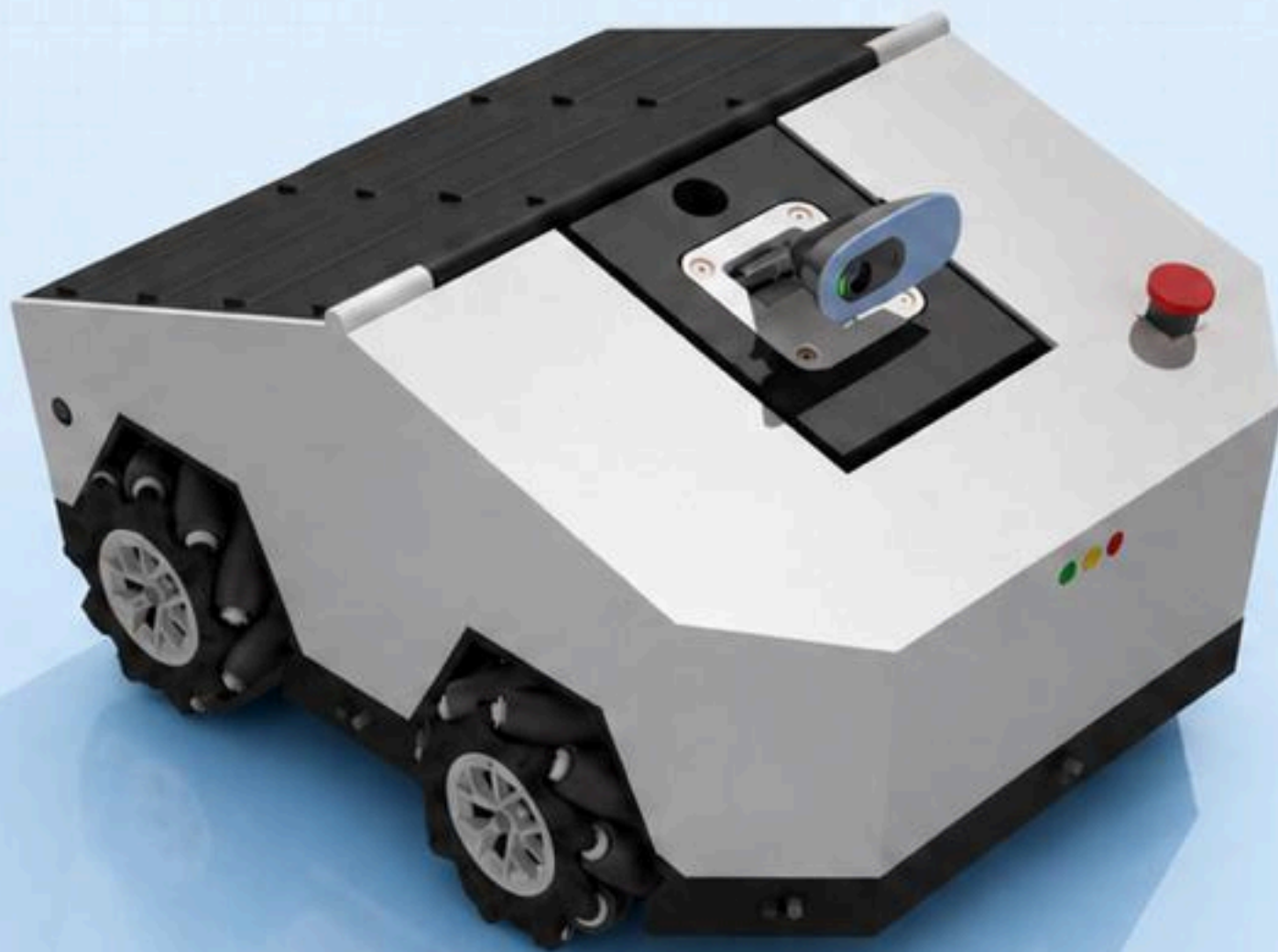


Check for loose screws and damaged parts.



If the issue persists, contact your instructor or support team.

TECHNICAL SPECIFICATIONS



1. MECHANICAL

Overall Dimensions	385 × 330 × 170 mm
Mass	5 kg
Body Material	3D-printed PLA
Wheel Configuration	4 mecanum wheels
Drive Motors	DC motors, 60 RPM
Lift Actuator	NEMA 17 stepper motor



2. ELECTRICAL & CONTROL

Battery	12V Li-ion battery, 10,000 mAh
Main Controller	Raspberry Pi 5
Low-Level Controller	ESP32 Development Kit
Camera Module	Logitech camera module
IMU	MPU6050



3. PROTECTION & SAFETY

Overcurrent Protection	Fuse
Reverse Polarity Protection	Included
Emergency Stop (E-Stop)	Cuts actuator power only
Buzzer Function	Activates when an obstacle is detected



4. STATUS INDICATORS

Green Indicator	Ready
Yellow Indicator	Moving
Red Indicator	Scanning

AUDIX

SCAN. VERIFY. REPORT.

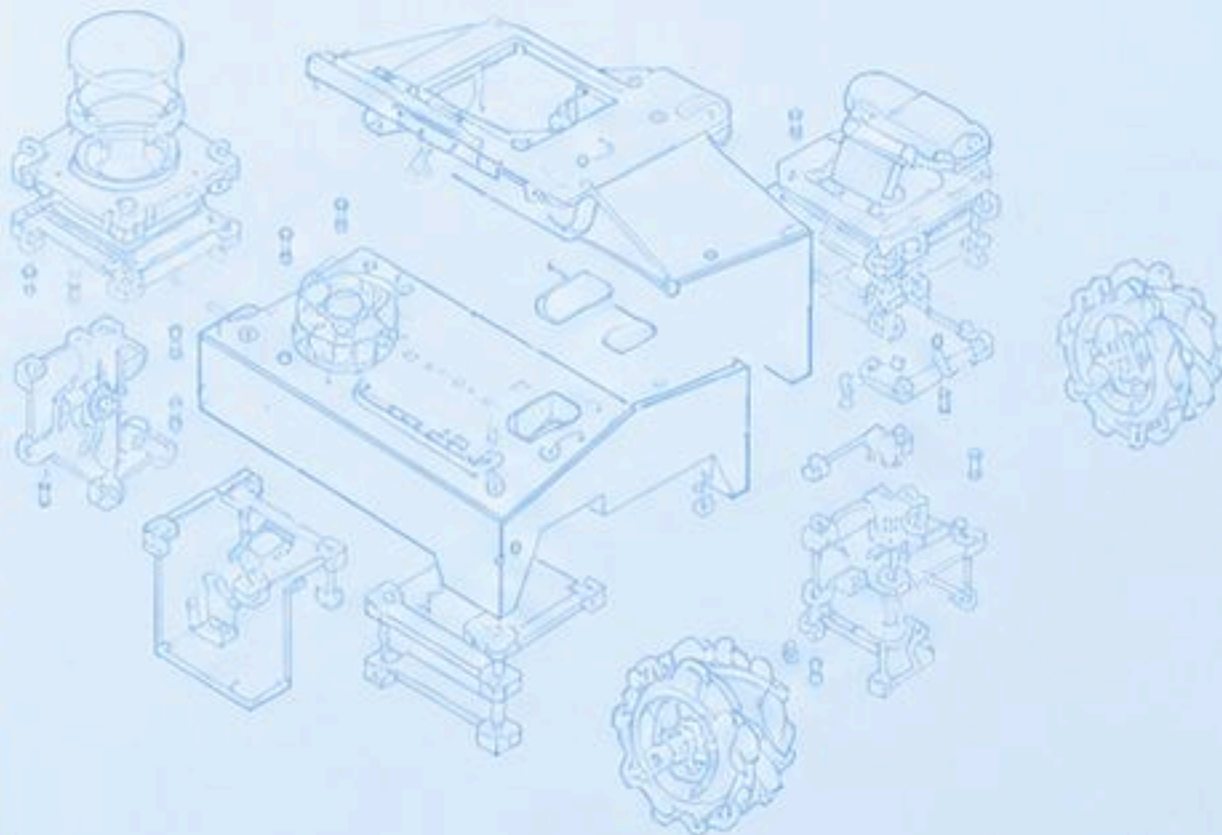
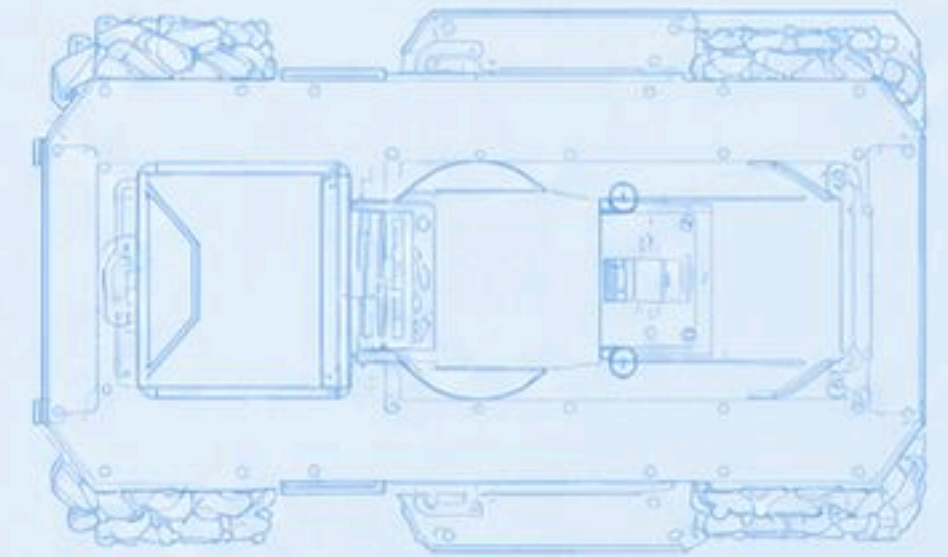
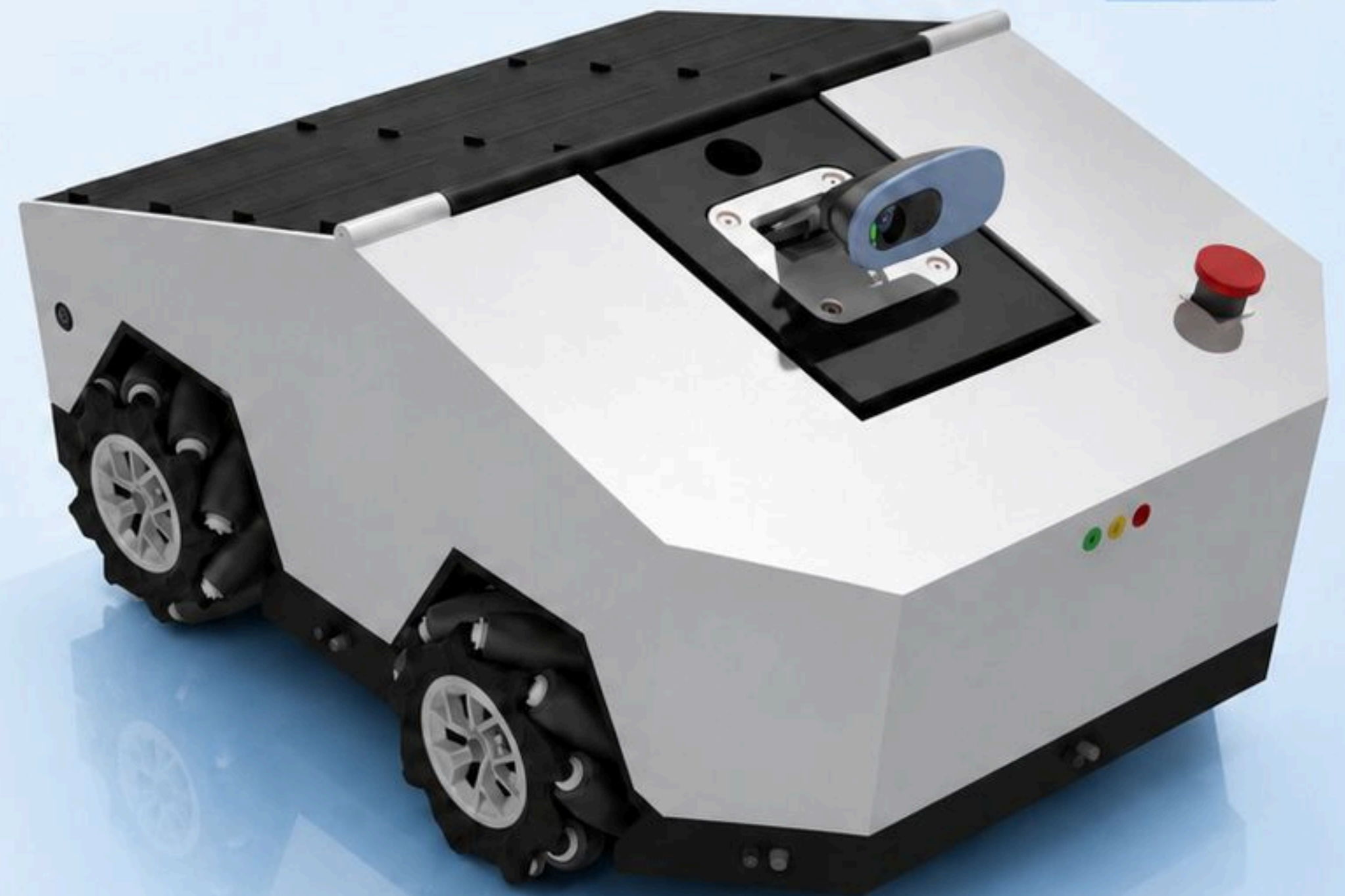
Thank you for being
part of **the future.**



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